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Methods

# Debiasing In-Sample Policy Performance for Small-Data, Large-Scale Optimization

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**Abstract.** Motivated by the poor performance of cross-validation in settings where data are scarce, we propose a novel estimator of the out-of-sample performance of a policy in data-driven optimization. Our approach exploits the optimization problem's sensitivity analysis to estimate the gradient of the optimal objective value with respect to the amount of noise in the data and uses the estimated gradient to debias the policy's in-sample performance. Unlike cross-validation techniques, our approach avoids sacrificing data for a test set and uses all data when training and hence is well suited to settings where data are scarce. We prove bounds on the bias and variance of our estimator for optimization problems with uncertain linear objectives but known, potentially nonconvex, feasible regions. For more specialized optimization problems where the feasible region is "weakly coupled" in a certain sense, we prove stronger results. Specifically, we provide explicit high-probability bounds on the error of our estimator that hold uniformly over a policy class and depends on the problem's dimension and policy class's complexity. Our bounds show that under mild conditions, the error of our estimator vanishes as the dimension of the optimization problem grows, even if the amount of available data remains small and constant. Said differently, we prove our estimator performs well in the small-data, large-scale regime. Finally, we numerically compare our proposed method to state-of-the-art approaches through a case-study on dispatching emergency medical response services using real data. Our method provides more accurate estimates of out-of-sample performance and learns better-performing policies.

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**Keywords:** small-data, large-scale • data-driven optimization • large-scale regime • cross-validation • end-to-end optimization

## 1. Introduction

The crux of data-driven decision making is using past data to identify decisions that will have good out-of-sample performance on future unseen data. Indeed, estimating out-of-sample performance is key to both policy evaluation (assessing the quality of a given policy) and to policy learning (identifying the best policy from a potentially large set of candidates). Estimating out-of-sample performance, however, is nontrivial. Naive estimates that leverage the same data to train a policy and to evaluate its performance often suffer a systematic, optimistic bias, referred to as "in-sample bias" in machine learning and the "optimizer's curse" in optimization (Smith and Winkler 2006).

Consequently, cross-validation and sample-splitting techniques have emerged as the gold-standard approach to estimating out-of-sample performance. Despite the multitude of cross-validation methods, at a high level, these methods all proceed by setting aside a portion of the data as "testing" data *not* to be used when training the policy and then evaluating the policy on these testing data. The policy's performance on testing data then serves as an estimate of its performance on future, unseen data, thereby circumventing the previously mentioned in-sample bias. Cross-validation is ubiquitous in machine learning and statistics with provably good performance in large sample settings (Kearns and Ron 1999, Bousquet and Elisseeff 2001).

Unfortunately, when data are scarce, cross-validation can perform poorly. Gupta and Rusmevichientong (2021) prove that for the small-data, large-scale regime—when the number of uncertain parameters in an optimization problem is large but the amount of relevant data per parameter is small—each of hold-out, 5-fold, 10-fold, and

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leave-one-out cross-validation can have poor performance when used for policy learning, even for very simple optimization problems. Shao (1993) observes a similar failure for leave-one-out cross-validation in a high-dimensional linear regression setting. The key issue in both cases is that when relevant data are *scarce*, estimates of uncertain parameters are necessarily imprecise, and omitting even a small amount of data when training a policy dramatically degrades its performance. Hence, the performance of a policy trained *with a portion of the data* on the test set is not indicative of the performance of the policy trained *with all the data* on future unseen data. We elucidate this phenomenon with a stylized example in Section 1.2.

Worse, this phenomenon is not merely an intellectual curiosity. Optimization problems plagued by numerous low-precision estimates are quite common in modern, large-scale operations. For example, optimization models for personalized pricing necessarily include parameters for each distinct customer type, and these parameters can be estimated only imprecisely because relevant data for each type are limited. Similar issues appear in large-scale supply chain design, promotion optimization, and dispatching emergency response services (see Section 2 for further discussion).

In this paper, we propose a new method for estimating out-of-sample performance without sacrificing data for a test set. The key idea is to debias the in-sample performance of the policy trained on all the data. Specifically, we focus on the optimization problem:

$$x^* \in \arg\min_{x \in \mathcal{X} \subseteq [0,1]^n} \quad \boldsymbol{\mu}^\top x, \tag{1.1}$$

where $\mathcal{X}$ is a known, potentially nonconvex feasible region contained within $[0,1]^n$, and $\boldsymbol{\mu} \in \mathbb{R}^n$ is an unknown vector of parameters. We assume access to a vector $\boldsymbol{Z}$ of noisy, unbiased predictions of $\boldsymbol{\mu}$ (based on historical data) and are interested in constructing a policy $\boldsymbol{x}(\boldsymbol{Z})$ with good out-of-sample performance $\boldsymbol{\mu}^\top \boldsymbol{x}(\boldsymbol{Z})$. (For clarity, the in-sample performance of $\boldsymbol{x}(\boldsymbol{Z})$ is $\boldsymbol{Z}^\top \boldsymbol{x}(\boldsymbol{Z})$.) For many applications of interest, $\boldsymbol{\mu}^\top x^* = O(n)$ as $n \to \infty$; that is, the full-information solution grows at least linearly as the dimension grows. Hence, the unknown out-of-sample performance $\boldsymbol{\mu}^\top \boldsymbol{x}(\boldsymbol{Z})$ must also be at least $O_p(n)$ as $n \to \infty$.[1] See Section 2 for examples.

Despite its simplicity, Problem (1.1) subsumes a wide class of optimization problems because $\mathcal{X}$ can be nonconvex and/or discrete. This class includes mixed-binary linear optimization problems such as facility location, network design, and promotion maximization. By transforming decision variables, even some nonlinear optimization problems such as personalized pricing can be rewritten as Problem (1.1) (see Section 2). In this sense, Problem (1.1) is fairly general.

Our estimator applies to classes of affine plug-in policies that are formally defined in Section 2. Loosely, affine plug-in policies are those obtained by solving Problem (1.1) after "plugging-in" some estimator $r_j(Z_j)$ in place of $\mu_j$, and $r_j(Z_j)$ depends affinely on $Z_j$. Many policies used in practice and previously studied in the literature can be viewed as elements of an affine plug-in policy class including sample average approximation (SAA), estimate-then-optimize policies based on regression, the Bayes-inspired policies of Gupta and Rusmevichientong (2021), and the Smart "Predict, then Optimize" (SPO+) policy of Elmachtoub and Grigas (2022). Thus, our estimator provides a theoretically rigorous approach to assessing the quality of optimization policies based on many modern machine learning techniques.

We debias $\boldsymbol{Z}^\top \boldsymbol{x}(\boldsymbol{Z})$ by exploiting the structure of Problem (1.1) with the plug-in $\boldsymbol{r}(\boldsymbol{Z})$. Specifically, by leveraging this problem's sensitivity analysis, we approximately compute the gradient of its objective value with respect to the variance of $\boldsymbol{Z}$ and use the estimated gradient to debias the in-sample performance. We term this correction the *variance gradient correction* (VGC). Because our method strongly exploits optimization structure, the VGC is Lipschitz continuous in the plug-in values $\boldsymbol{r}(\boldsymbol{Z})$. This continuity is not enjoyed by other techniques such as those in Gupta and Rusmevichientong (2021).

Although the VGC's continuity may seem like mere a mathematical nicety, empirical evidence suggests it improves empirical performance. Similar empirical phenomena, where an estimator that varies smoothly in the data often outperforms similar estimators that change discontinuously, are rife in machine learning. We see such phenomenon when comparing $k$-nn with Gaussian kernel smoothing (Friedman et al. 2001), Classification and Regression Trees (CART) with bagged trees (Breiman 1996), or best subset-regression with least absolute shrinkage and selection operator (LASSO) regression (Hastie et al. 2020a). Theoretically, we exploit this smoothness heavily to establish bounds that hold uniformly over the policy class.

Specifically, we show that, when $\boldsymbol{Z}$ is approximately Gaussian, the bias of our estimator for out-of-sample performance is $\tilde{O}(h)$ as $h \to 0$, where $h$ is a user-defined parameter that controls the accuracy of our gradient estimate (Theorem 3.2). Characterizing the variance is more delicate. We introduce the concept of *average solution instability* and prove that if the instability of the policy vanishes at rate $O(n^{-\alpha})$ for $\alpha \geq 0$, then the variance of our estimator is roughly $O(n^{3-\alpha}/h)$. Collectively, these results suggest interpreting $h$ as a parameter controlling the bias-variance tradeoff of our estimator. Moreover, when $\alpha > 1$, the variance of our estimator is $o(n^2)$. Because, as mentioned, the unknown out-of-sample performance often grows at least linearly in $n$, that is, $\boldsymbol{\mu}^\top \boldsymbol{x}(\boldsymbol{Z}) = O_p(n)$, our variance bound

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shows that when $\alpha > 1$ and $n$ is large, the stochastic fluctuations of our estimator are negligible relative to the out-of-sample performance. In other words, our estimator is quite accurate in these settings.

Our notion of average solution instability is formally defined in Section 3.3. Loosely, it measures the expected change in the $j$th component of the policy after replacing the $k$th data point with an independent and identically distributed (i.i.d.) copy, where $j$ and $k$ are chosen uniformly at random from $\{1, \ldots, n\}$. This notion of stability is similar to hypothesis stability (Bousquet and Elisseeff 2001), but, to the best of our knowledge, is distinct. Moreover, insofar as we expect that a small perturbation of the data are unlikely to have a large change on the solution for most real-world, large-scale optimization problems, we expect average solution instability to be small and our estimator to have low variance.

We then prove stronger high-probability tail bounds on the error of our estimator for two special classes of "weakly coupled" instances of Problem (1.1): weakly coupled-by-variables and weakly coupled-by-constraints. In Section 4.1, we consider problems that are weakly coupled-by-variables, that is, problems that decouple into many, disjoint subproblems once a small number of decision variables are fixed. In Section 4.2, we consider problems that are weakly coupled-by-constraints, that is, problems that decouple into many, disjoint subproblems once a small number of constraints are removed. For each problem class, we go beyond bounding the variance to provide an explicit tail bound on the relative error of our estimator that holds uniformly over the policy class. We show that for problems weakly coupled-by-variables the relative error scales like $\tilde{O}(C_{\mathsf{PI}}\ (\mathrm{polylog}(1/\epsilon))/\sqrt[3]{n})$, where $C_{\mathsf{PI}}$ is a constant measuring the complexity of the policy class (see Theorem 4.3). Similarly, we show the relative error for problems weakly coupled-by-constraints scales like $\tilde{O}(C_{\mathsf{PI}}(\mathrm{polylog}(1/\epsilon))/\sqrt[4]{n})$, where $C_{\mathsf{PI}}$ measures both the complexity of the policy class and number of constraints of the problem (see Theorem 4.7). Importantly, because these bounds hold uniformly, our debiased in-sample performance can be used both for policy evaluation and policy learning, even when data are scarce, as long as $n$ (the dimension of the problem) is sufficiently large. Said differently, our estimator of out-of-sample performance is particularly well suited to small-data, large-scale optimization.

Admittedly, weakly coupled problems as described previously do not cover all instances of Problem (1.1), and the appropriateness of modeling $\boldsymbol{Z}$ as approximately Gaussian is application specific. Nonetheless, our results and their proofs highly suggest our estimator will have strong performance whenever the underlying optimization problem is well behaved enough for certain uniform laws of large numbers to pertain.

Finally, to complement these theoretical results, we perform a numerical case study of dispatching emergency medical services with real data from cardiac arrest incidents in Ontario, Canada. With respect to policy evaluation and learning, we show that our debiased in-sample performance outperforms both traditional cross-validation methods and the Stein correction of Gupta and Rusmevichientong (2021). In particular, although the bias of cross-validation is nonvanishing as the problem size grows for a fixed amount of data, the bias of our VGC converges to zero. Similarly, although both the Stein correction and our VGC have similar asymptotic performance, the smoothness of VGC empirically leads to lower bias and variance for moderately sized instances.

### 1.1. Our Contributions

We summarize our contributions as follows:

1. We propose an estimator of out-of-sample performance for Problem (1.1) by debiasing in-sample performance through a novel VGC. Our VGC applies to a general class of affine plug-in policies that subsumes many policies used in practice. Most importantly, unlike cross-validation, VGC does *not* sacrifice data when training, and hence is particularly well suited to settings where data are scarce.

2. We prove that under some assumptions on the data-generating process, for general instances of Problem (1.1), the bias of our estimator is at most $\tilde{O}(h)$ as $h \to 0$, where $h$ is a user-defined parameter. For policy classes that satisfy a certain average solution instability condition, we also prove that that its variance scales like $o\left(\frac{n^2}{h}\right)$ as $n \to \infty$.

3. We prove stronger results for instances of Problem (1.1) in which the feasible region is only weakly coupled. When the feasible region is weakly coupled by variables, we prove that, with probability at least $1-\epsilon$, debiasing in-sample performance with our VGC recovers the true out-of-sample performance up to relative error that is at most $\tilde{O}\left(C_{\mathsf{PI}}\frac{\log(1/\epsilon)}{n^{1/3}}\right)$ as $n \to \infty$, uniformly over the policy class, where $C_{\mathsf{PI}}$ is a constant that measures the complexity of the plug-in policy class (Theorem 4.3). Similarly, for certain linear optimization problems that are weakly coupled by constraints, we prove that, with probability at least $1-\epsilon$, debiasing in-sample performance with VGC estimates the true out-of-sample performance uniformly over the policy class with relative error that is at most $\tilde{O}\left(C_{\mathsf{PI}}\sqrt{\log(1/\epsilon)}/n^{1/4}\right)$, where $C_{\mathsf{PI}}$ is a constant measuring the complexity of the plug-in policy class and the number of constraints of the problem (Theorem 4.7). We stress that because both these bounds hold uniformly, our debiased in-sample performance cannot only be used for policy evaluation, but also policy

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learning, even when data are scarce, so long as $n$ (the size of the problem) is sufficiently large.

4. Finally, we present a numerical case study based on real data from dispatching emergency response services to highlight the strengths and weaknesses of our approach relative to cross-validation and the Stein correction of Gupta and Rusmevichientong (2021). Overall, we find that because our VGC exploits the optimization structure of Problem (1.1), it outperforms the benchmarks when the number of uncertain parameters is sufficiently large. Additionally, in settings where the signal to noise ratio is low, VGC more effectively balances the bias-variance tradeoff than cross-validation, which can be quite sensitive to the number of folds used.

### 1.2. Motivating Example: Poor Performance of Cross-Validation with Limited Data

Before proceeding, we present an example that highlights the shortcomings of cross-validation and the benefits of our method when data are limited. Consider a special case of Problem (1.1):

$$\max_{x\in\{0,1\}^n} \sum_{j=1}^{n} \mu_j x_j, \tag{1.2}$$

where the true parameters $\boldsymbol{\mu} \in \{-1,1\}^n$ are unknown, but we observe $S$ samples $\boldsymbol{Y}_1,\ldots,\boldsymbol{Y}_S$, where $\boldsymbol{Y}_i \in \mathbb{R}^n$ and $\boldsymbol{Y}_i \sim \mathcal{N}(\boldsymbol{\mu}, 2\boldsymbol{I})$ for all $i$ and $\boldsymbol{I}$ is the identity matrix. A standard data-driven policy in this setting is SAA, also called empirical risk minimization, which prescribes the following policy:

$$\boldsymbol{x}^{\mathsf{SAA}}(\boldsymbol{Z}) \in \arg\max_{x\in\{0,1\}^n} \sum_{j=1}^{n} Z_j x_j \qquad \text{where} \qquad Z_j = \frac{1}{S}\sum_{i=1}^{S} Y_{ij}.$$

The key question, of course, is "what is SAA's out-of-sample behavior $\boldsymbol{\mu}^\top \boldsymbol{x}^{\mathsf{SAA}}(\boldsymbol{Z})$?"

To study this question, the left panel of Figure 1 shows different estimators for the out-of-sample performance of SAA $\mathbb{E}[\boldsymbol{\mu}^\top \boldsymbol{x}^{\mathsf{SAA}}(\boldsymbol{Z})]$ when $S = 3$ in Problem (1.2). The right panel shows the expected relative error (with respect to the oracle) of these estimators as the number of samples $S$ grows. To account for the noise level of the samples, we plot the estimation error with respect to the signal-to-noise ratio (SNR) of $Z_j$.[2] For reference, Hastie et al. (2020b) argues that SNR greater than one is rare when working with noisy, observational data, and an SNR of 0.25 is more typical.

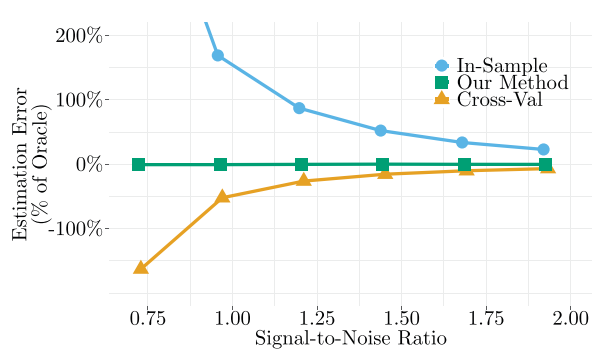
The first row of Figure 1 presents the in-sample performance, that is, the objective of the SAA problem. As expected, we see in-sample performance significantly overestimates the out-of-sample performance. The right panel of Figure 1 suggests this effect persists across SNRs and the relative error is at least 23% for SNRs less than two.

The second row of the left panel of Figure 1 shows the leave-one-out cross-validation error, which aims to correct the overoptimistic bias and computes $\frac{1}{S}\sum_{i=1}^{S} \boldsymbol{Y}_i^\top \boldsymbol{x}^{\mathsf{SAA}}(\boldsymbol{Z}^{-i})$, where $\boldsymbol{Z}^{-i} = \frac{1}{S-1}\sum_{j\neq i} \boldsymbol{Y}_j$. Cross-validation is also fairly inaccurate, suggesting SAA performs worse than the trivial, non–data-driven policy $\boldsymbol{x} = \boldsymbol{0}$, which has an out-of-sample performance of zero. In the right panel, this incorrect implication occurs for SNRs less than about 0.875.

Why does cross-validation perform so poorly? *By construction*, cross-validation omits some data in training and hence does *not* estimate the out-of-sample performance of $\boldsymbol{x}^{\mathsf{SAA}}(\boldsymbol{Z})$, but rather, that of $\boldsymbol{x}^{\mathsf{SAA}}(\boldsymbol{Z}^{-1})$.[3] From the second column of Figure 1, we see the cross-validation estimate *does* nearly match the true (oracle) performance of $\boldsymbol{x}^{\mathsf{SAA}}(\boldsymbol{Z}^{-1})$. When data are scarce, sacrificing even a small amount of data in training can dramatically degrade a policy. As seen in the right panel of Figure 1, this phenomenon is nonnegligible (at least 10% relative error) for SNRs less than or equal to 1.75.

**Figure 1.** (Color online) Expected Estimates of Out-of-Sample Performance by Policy for Problem (1.2)

| | $\boldsymbol{x}^{\mathsf{SAA}}(\boldsymbol{Z})$ | $\boldsymbol{x}^{\mathsf{SAA}}(\boldsymbol{Z}^{-i})$ |
|---|---|---|
| In-Sample | 18.36 | 22.33 |
| Cross-Val | −1.86 | −9.98 |
| Our Method | 2.95 | −1.89 |
| Oracle | 2.97 | −1.87 |




*Notes.* In the left table, we take $n = 100$, $S = 3$, and $\mu_j = 1$ if $j \le 14$ and $\mu_j = -1$ otherwise. We estimate the expected out-of-sample performance across 1,000,000 simulations. Standard errors are less than 0.005. In the right graph, we plot the bias of the estimates with respect to the expected out-of-sample performance as we increase the signal-to-noise ratio. The in-sample error not shown at 0.75 SNR exceeds 500%.

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Thus, the performance of $\boldsymbol{x}^{\mathrm{SAA}}(\boldsymbol{Z}^{-1})$ may not always be a good proxy of the performance of $\boldsymbol{x}^{\mathrm{SAA}}(\boldsymbol{Z})$.

How then might we resolve the issue? The third row of the left panel of Figure 1 presents our estimator based on debiasing the in-sample performance of $\boldsymbol{x}^{\mathrm{SAA}}(\boldsymbol{Z})$ with our VGC. Our estimate is essentially unbiased (see also Theorem 3.2). The right panel of Figure 1 confirms this excellent behavior across a range of SNRs. Finally, although this example focuses on the bias of our estimator, our results in Section 4 are stronger and bound the (random) error of our estimator directly rather than its expectation.

### 1.3. Relationship to Prior Work

Cross-validation is the gold standard for estimating out-of-sample performance in the large-sample regime with i.i.d. data (see Bousquet and Elisseeff (2001) and Kearns and Ron (1999) for some fundamental results). As discussed previously, when estimating the performance of a fixed-policy, these approaches entail sacrificing some data in the training step to set aside for validation and hence may be ill suited to data-scarce settings. Similar issues arise in a variety of other sample-splitting methods, including "honest-trees" Wager and Athey (2018) and most forms of doubly robust estimation Dudík et al. (2011). By contrast, our VGC-based approach to debiasing the in-sample performance effectively uses all the data when training, making it somewhat better suited to data-scarce settings and small-data, large-scale optimization.

Our work also contributes to the growing literature on "optimization-aware" estimation. These works use a variety of names including operational statistics (Liyanage and Shanthikumar 2005), learning-enabled optimization (Liu et al. 2022), decision-focused learning (Wilder et al. 2019a), end-to-end learning (Wilder et al. 2019b) and task-based learning (Donti et al. 2017). Fundamentally, this area of research seeks estimators that optimize the out-of-sample performance of a policy in a downstream optimization problem rather than the prediction error of the estimate. Closest to our work is the "smart 'predict then optimize'" framework studied in Elmachtoub and Grigas (2022) and Elmachtoub et al. (2020). These works also study Problem (1.1) but in a slightly different data setting and propose policy selection methods for affine and tree-based policies, respectively. Also related is Ito et al. (2018), who develop an unbiased estimate of the SAA policy for Problem (1.1) but do not consider higher level moments, policy evaluation for other policies, or policy learning.

Recently, Elmachtoub et al. (2020) and Hu et al. (2022) have sought to establish generalization guarantees for data-driven policies for plug-in policies for Problem (1.1), that is, bounds on the difference between in-sample performance and out-of-sample performance that hold uniformly over the policy class. Both works prove generalization guarantees that vanish in the large-sample regime (when the amount of data grows large). We similarly bound the difference between our *debiased* in-sample performance and out-of-sample performance, uniformly over a policy class. However, unlike the previous works, our bounds are specifically constructed to vanish relative to the unknown out-of-sample performance in the small-data, large-scale regime. When applied to the types of policies studied in Elmachtoub et al. (2020) and Hu et al. (2022), our debiased in-sample performance equals the ordinary in-sample performance (see discussion of "Policy Classes That Do Not Depend on $\boldsymbol{Z}$" in Section 3). Hence, our results can be reinterpreted as generalization guarantees for these classes, showing that generalization error vanishes relative to the out-of-sample performance as the problem size grows. In this sense, our work complements the large-sample analysis of Elmachtoub et al. (2020) and Hu et al. (2022) with new, small-data, large-scale analysis.

More generally, there has been somewhat less work in the small-data, large-scale regime, most notably Gupta and Kallus (2022) and Gupta and Rusmevichientong (2021). Of these, Gupta and Rusmevichientong (2021) is most closely related to our work. Loosely, Gupta and Rusmevichientong (2021) study a class of weakly coupled linear optimization problems and propose an estimator of the out-of-sample performance based on Stein's lemma. By leveraging a careful duality argument, the authors prove that the estimation error of their procedure vanishes in both the large-sample and small-data, large-scale regime.

Our work differs in two important respects: First, our estimator applies to a more general class of problems and more general policy classes. Indeed, we focus on Problem (1.1) with specialized results for weakly coupled instances. Our weakly coupled by constraints variant in Section 4.2 mirrors the setting of Gupta and Rusmevichientong (2021), and our weakly coupled by variables variant in Section 4.1 is more general, allowing us to model, for example, discrete optimization problems. Moreover, our affine plug-in policy class significantly generalizes the Bayes-inspired policy class of Gupta and Rusmevichientong (2021) by incorporating covariate information.

The second important difference from Gupta and Rusmevichientong (2021) relates to exploiting optimization structure in Problem (1.1). Gupta and Rusmevichientong (2021) fundamentally relies on Stein's lemma, a result that applies to general functions and does not specifically leverage optimization structure. By contrast, our method directly leverages the structure of Problem (1.1) through its sensitivity analysis and Danskin's theorem. By leveraging optimization structure, our VGC is,

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by construction, continuous in the policy class. Evidence from Section 5 suggests this smoothness yields an empirical advantage of our method.

Finally, our work also contributes to a growing literature on debiasing estimates in high-dimensional statistics, most notably for LASSO regression (Zhang and Zhang 2014, Javanmard and Montanari 2018) and *M*-estimation (Javanmard and Montanari 2014). Like these works, VGC involves estimating a gradient of the underlying system and using this gradient information to form a correction. Unlike these works, however, our gradient estimation strongly leverages ideas from sensitivity analysis in optimization. Moreover, the proofs of our performance guarantees involve substantively different mathematical techniques.

## 2. Model

As mentioned, our focus is on data-driven instances of Problem (1.1) where the feasible region $\mathcal{X}$ is known, but the parameters $\boldsymbol{\mu}$ are unknown. Despite its simplicity, several applications can be modeled in this form after a suitable transformation of variables.

**Example 2.1** (Promotion Optimization). Promotion optimization is an increasingly well-studied application area (Cohen et al. 2017, Baardman et al. 2019). Our formulation mirrors a formulation from the ride-sharing company Lyft around incentive allocation (Schmoys and Wang 2019) but also resembles the online advertising portfolio optimization problem (Rusmevichientong 2006, Pani and Sahin 2017, Gupta and Rusmevichientong 2021).

The decision maker (platform) seeks to allocate $J$ different types of coupons (promotions) to $K$ different customer (passenger) types. Coupons are costly, and there is a finite budget $C$ available. Let $\mu_{jk}$ be the reward (induced spending) and $c_{jk}$ be the cost of assigning coupon type $j$ to customer type $k$. Using $x_{jk}$ to denote the fraction of customers of type $k$ who receive coupons of type $j$, we can formulate the following linear optimization problem of the form of Problem (1.1):

$$\max_{x\geq 0}\left\{\sum_{k=1}^{K}\sum_{j=1}^{J}\mu_{jk}x_{jk} \;:\; \sum_{j=1}^{J}x_{jk}\leq 1 \quad \text{for each } k=1,\ldots,K,\right.$$
$$\left.\sum_{k=1}^{K}\sum_{j=1}^{J}c_{jk}x_{jk}\leq C\right\}.$$

In typical instances, the cost $c_{jk}$ are likely known (a "\$10 off" coupon costs \$10), whereas the reward $\mu_{jk}$ must be estimated from historical data. In settings with many types of coupons and customers, we might further expect that the reward estimates may be imprecise.

Some reflection suggests many linear optimization problems including shortest path with uncertain edge costs or even binary linear optimization problems like multichoice knapsack with uncertain rewards can be cast as shown previously.

We next observe that some two-stage linear optimization problems can also be framed as Problem (1.1).

**Example 2.2** (Drone-Assisted Emergency Medical Response). In recent years, emergency response systems have begun using drones as part of their operations, specifically for rapid delivery of automatic electronic defibrillators (AEDs) for out-of-hospital cardiac arrests (OHCA) (Sanfridsson et al. 2019, Cheskes et al. 2020). The intention is that a drone might reach a patient in a remote region before a dispatched ambulance, and (untrained) bystanders can use the AED to assist the patient until the ambulance arrives. Consequently, researchers have begun studying both how to design a drone-assisted emergency response network (where to locate depots) (Boutilier and Chan 2022) and how to create optimal dispatch rules (to which locations should we allocate a drone and from which depot) (Chu et al. 2021). Combining these two problems yields a two-stage optimization problem, similar to facility location, aimed at minimizing the response time.

Namely, let $\mu_{kl}$ be the response time of drone routed from a source $l$ for to a patient location $k$, $l=1,\ldots,L$, and $k=1,\ldots,K$. Let $y_l\in\{0,1\}$ be a binary decision variable encoding if we build a drone depot at location $l$, and let $x_{kl}$ be a binary decision variable encoding if, after building the network, we should dispatch a drone from location $l$ to patient requests at location $k$. We let $x_{k0}$ be the choice not to route a drone (sending only an ambulance) to location $k$ and $\mu_{k0}$ be the corresponding ambulance travel time. Suppose we can build at most $B$ depots. Then, we have the following optimization problem:

$$\min_{y\in\{0,1\}^L,\, x\in\{0,1\}^{K\times L}} \quad \sum_{k=1}^{K}\sum_{l=0}^{L}\mu_{kl}x_{kl},$$
$$\text{s.t.} \quad \sum_{l=1}^{L}y_l\leq B, \quad x_{kl}\leq y_l, \quad \sum_{l=0}^{L}x_{kl}=1,$$
$$\forall k=1,\ldots,K,\ l=1,\ldots,L.$$

Insofar as some drone response times are difficult to predict (depending on the weather, local environment, ability of bystanders to locate and use the drone's payload), we expect in typical instances that estimates $\mu_{kl}$ may be imprecise.

Interestingly, some nonlinear problems can be transformed into the form of Problem (1.1).

**Example 2.3** (Personalized Pricing). Personalized pricing strategies seek to assign a tailored price to each of many customer types reflecting their heterogeneous willingness-to-pay (Aouad et al. 2019, Javanmard et al.

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2020, Cohen et al. 2021). One simple formulation posits distinct demand models $D_j(p) = m_j\phi_j(p) + b_j$ in each customer segment $j$, for some decreasing function $\phi_j(p)$. This yields the revenue maximization problem:

$$\max_{p\geq 0} \quad \sum_{j=1}^{n} m_j p_j \phi_j(p_j) + b_j p,$$

where $p_j$ is the price for the $j$th segment. We can cast this nonlinear objective in the form Problem (1.1) by transforming variables:

$$\max_{p\geq 0, x} \left\{ \sum_{j=1}^{n} m_j x_j + b_j p_j \ : \ x_j = p_j \phi_j(p_j) \quad \text{for each } j = 1, \ldots, n \right\},$$

where the resulting feasible region is now nonconvex. In typical settings, we expect the parameters $m_j$ and $b_j$ are unknown and estimated via machine learning methods (Aouad et al. 2019). When there are many customer types, these estimates may be imprecise for rarely occurring types.

Finally, we mention as an aside that some dynamic programs like the economic lot-sizing problem in inventory management can be cast in the above form through a careful representation of the dynamic program (see Elmachtoub and Grigas (2022) for details).

### 2.1. Data

Following Gupta and Rusmevichientong (2021), we do not assume an explicit data generation procedure. Rather, we assume that as a result of analyzing whatever raw data are available, we obtain noisy, unbiased predictions $\boldsymbol{Z}$ such that $\mathbb{E}[\boldsymbol{Z}] = \boldsymbol{\mu}$ with known precision $\mathbb{E}[(Z_j - \mu_j)^2] = 1/\nu_j$ for $j = 1, \ldots, n$. These predictions might arise as sample averages as in Section 1.2 or as the outputs of some preprocessing regression procedure. We further assume that for each $j = 1, \ldots, n$, we observe a nonrandom covariate of feature data $\boldsymbol{W}_j \in \mathbb{R}^p$, which may (or may not) be informative for the unknown $\mu_j$.

We believe this setup reasonably reflects many applications. In the case of drone-assisted emergency response (Example 2.2), $\boldsymbol{W}_j$ encodes features that are predictive of Emergency Medical Services response times such as physical road distance between the patient and the responding ambulance, time of day, day of week, and weather conditions (Chu et al. 2021), whereas $Z_{k0}$ may be an average of historical response times to location $k$.

An advantage of modeling $\boldsymbol{Z}$ in lieu of the data generation process is that the precisions $\nu_j$ implicitly describe the amount of relevant data available for each $\mu_j$. Let $\nu_{\min} \equiv \min_j \nu_j$ and $\nu_{\max} \equiv \max_j \nu_j$. Then, loosely speaking, the large-sample regime describes instances where $\nu_{\min}$ is large, that is, where data are plentiful and we can estimate $\boldsymbol{\mu}$ easily. By contrast, the small-data, large-scale regime describes instances in which $n$ is large (large-scale), but there are limited relevant data, and hence $\nu_{\max}$ is small.

To simplify our exposition, we will also assume the following.

**Assumption 2.4** (Independent Gaussian Corruptions). *For each $j = 1, \ldots, n$, $Z_j$ has Gaussian distribution with $Z_j \sim \mathcal{N}(\mu_j, 1/\nu_j)$ where $\nu_j$ is the known precision of $Z_j$. Moreover, $Z_1, \ldots, Z_n$ are independent.*

Assumption 2.4 is common. Gupta and Rusmevichientong (2021) use a similar assumption; Javanmard and Montanari (2018) strongly leverage a Gaussian design assumption when debiasing LASSO estimates; and Ito et al. (2018) also assumes Gaussian errors in their debiasing technique. In each case, the idea is if a technique enjoys *provably* good performance under Gaussian corruptions, it will likely have good *practical* performance when data are approximately Gaussian. Indeed, if $\boldsymbol{Z}$ is obtained by maximum likelihood estimation, ordinary linear regression, simple averaging as in Section 1.2, or Gaussian process regression, then the resulting estimates will be approximately Gaussian. We adopt the same perspective in our work.

The independence assumption in Assumption 2.4 is without loss of generality as illustrated in the following example.

**Example 2.5** (Correlated Predictions). Suppose we are given an instance of Problem (1.1) and predictions $\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, where $\Sigma$ is a *known*, positive semidefinite matrix. Consider a Cholesky decomposition $\Sigma = \boldsymbol{L}\boldsymbol{L}^\top$ and the transformed predictions $\overline{\boldsymbol{Z}} \equiv \boldsymbol{L}^{-1}\boldsymbol{Z}$. Notice $\overline{\boldsymbol{Z}} \sim \mathcal{N}(\overline{\boldsymbol{\mu}}, \boldsymbol{I})$, where $\overline{\boldsymbol{\mu}} \equiv \boldsymbol{L}^{-1}\boldsymbol{\mu}$. We then recast Problem (1.1) as the equivalent problem:

$$\min_{\overline{x}\in\overline{\mathcal{X}}} \quad \overline{\boldsymbol{\mu}}^\top \overline{x}, \quad \text{where} \quad \overline{\mathcal{X}} \equiv \{\boldsymbol{L}^\top \boldsymbol{x} \ : \ \boldsymbol{x} \in \mathcal{X}\}.$$

Our new problem is of the required form with transformed predictions independent across $j$.

Most importantly, Assumption 2.4 is *not* crucial to many of our results. Violating the Gaussian assumption only affects the bias of our estimator (see Theorem 3.2). Our analysis bounding the variance and tails of the stochastic errors use empirical process theory and can easily be adapted for non-Gaussian corruptions. Moreover, although the bias of our estimator is nonnegligible when $\boldsymbol{Z}$ is non-Gaussian, we can bound this bias in terms of the Wasserstein distance between $\boldsymbol{Z}$ and a multivariate Gaussian, suggesting our method has good performance as long as corruptions are *approximately* Gaussian and this distance is small (see Lemma B.4 in the online appendix for the bound).

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Finally, similar results also hold when $\boldsymbol{\nu}$ is, itself, estimated noisily with the addition of a small bias term related to its estimate's accuracy (see Lemma B.3 in the online appendix).

## 2.2. Affine Plug-in Policy Classes

A data-driven policy for Problem (1.1) is a mapping $\boldsymbol{Z} \longmapsto \boldsymbol{x}(\boldsymbol{Z}) \in \mathcal{X}$ that determines a feasible decision $\boldsymbol{x}(\boldsymbol{Z})$ from the observed data $\boldsymbol{Z}$. We focus on classes of affine plug-in policies. Intuitively, a plug-in policy first proxies the unknown $\boldsymbol{\mu}$ by some estimate, $\boldsymbol{r}(\boldsymbol{Z})$, and then solves Problem (1.1) after plugging-in this estimate for $\boldsymbol{\mu}$.

**Definition 2.6** (Affine Plug-in Policy Classes). For $j = 1, \ldots, n$, let $r_j(z, \boldsymbol{\theta}) = a_j(\boldsymbol{\theta})z + b_j(\boldsymbol{\theta})$ be an affine function of $z$ where $a_j(\boldsymbol{\theta})$ and $b_j(\boldsymbol{\theta})$ are arbitrary functions of the parameter $\boldsymbol{\theta} \in \Theta$. Let $\boldsymbol{r}(\boldsymbol{Z}, \boldsymbol{\theta}) = (r_1(Z_1, \boldsymbol{\theta}), r_2(Z_2, \boldsymbol{\theta}), \ldots, r_n(Z_n, \boldsymbol{\theta}))^\top \in \mathbb{R}^n$. The plug-in policy with respect to $\boldsymbol{r}(\cdot, \boldsymbol{\theta})$ is given by

$$\boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}) \in \arg\min_{x \in \mathcal{X}} \boldsymbol{r}(\boldsymbol{Z}, \boldsymbol{\theta})^\top \boldsymbol{x}, \tag{2.1}$$

where ties are broken arbitrarily. Furthermore, we let $\mathcal{X}_\Theta(\boldsymbol{Z}) \equiv \{\boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}) \in \mathcal{X} : \boldsymbol{\theta} \in \Theta\} \subseteq \mathcal{X}$ denote the corresponding class of plug-in policies over $\Theta$.

When $\boldsymbol{\theta}$ is fixed and clear from context, we suppress its dependence, writing $\boldsymbol{x}(\boldsymbol{Z})$ and $\boldsymbol{r}(\boldsymbol{Z})$. Moreover, for a fixed $\boldsymbol{\theta}$, $r_j(Z_j, \boldsymbol{\theta})$ only depends on the data (linearly) through the $j$th component.

Plug-in policies are attractive in data-driven optimization because computing $\boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta})$ involves solving a problem of the same form as Problem (1.1). Thus, if a specialized algorithm exists for solving Problem (1.1), for example, as with many network optimization problems, the same algorithm can be used to compute the policy. This property does not necessarily hold for other classes of policies such as regularization-based policies (Gupta and Rusmevichientong 2021).

Moreover, many policies used in practice are of the form $\boldsymbol{x}(\boldsymbol{Z}, \hat{\boldsymbol{\theta}}(\boldsymbol{Z}))$ for some $\hat{\boldsymbol{\theta}}(\boldsymbol{Z})$. Such policies are *not* affine plug-in policies; $r_j(Z_j, \hat{\boldsymbol{\theta}}(\boldsymbol{Z}))$ may depend nonlinearly on all the data $\boldsymbol{Z}$. Nonetheless, our analysis will bound the error of our estimator applied to such policies. Namely, in Section 4, we provide error bounds on our estimator that hold uniformly over $\mathcal{X}_\Theta(\boldsymbol{Z})$. Because these bounds hold uniformly, such bounds also hold for all policies of the form $\boldsymbol{x}(\boldsymbol{Z}, \hat{\boldsymbol{\theta}}(\boldsymbol{Z}))$.

For clarity, we make no claim about the optimality of affine plug-in policies for Problem (1.1); for a particular application, there may exist nonaffine policies with superior performance. Our focus on affine plug-ins is motivated by their ubiquity and computational tractability.

We next present examples:

• **SAA:** The SAA is a canonical data-driven policy for Problem (1.1). It is defined by

$$\boldsymbol{x}^{\mathsf{SAA}}(\boldsymbol{Z}) \in \arg\min_{x \in \mathcal{X}} \boldsymbol{Z}^\top \boldsymbol{x}. \tag{2.2}$$

SAA is thus an affine plug-in policy where the function $r_j(z, \boldsymbol{\theta}) = z$.

• **Plug-ins for Regression Models:** Consider the linear model $r_j(\boldsymbol{Z}, \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \boldsymbol{W}_j$, which does not depend on $\boldsymbol{Z}$, and the affine plug-in policy

$$\boldsymbol{x}^{\mathsf{LM}}(\boldsymbol{Z}, \boldsymbol{\theta}) \in \arg\min_{x \in \mathcal{X}} \sum_{j=1}^{n} \boldsymbol{W}_j^\top \boldsymbol{\theta} \cdot x_j. \tag{2.3}$$

As mentioned, many policies in the literature are of the form $\boldsymbol{x}^{\mathsf{LM}}(\boldsymbol{Z}, \hat{\boldsymbol{\theta}}(\boldsymbol{Z}))$ for a particular $\hat{\boldsymbol{\theta}}(\boldsymbol{Z})$. For example, letting $\boldsymbol{\theta}^{\mathsf{OLS}}(\boldsymbol{Z}) \in \arg\min_{\boldsymbol{\theta}} \sum_{j=1}^{n} (Z_j - \boldsymbol{\theta}^\top \boldsymbol{W}_j)^2$ be the ordinary least-squares fit yields the estimate-then-optimize policy $\boldsymbol{x}^{\mathsf{LM}}(\boldsymbol{Z}, \boldsymbol{\theta}^{\mathsf{OLS}}(\boldsymbol{Z}))$. Similarly, by appropriately padding the covariate with zeros, we can write the "optimization-aware" SPO and SPO+ methods of Elmachtoub and Grigas (2022) over linear hypothesis classes in the form $\boldsymbol{x}^{\mathsf{LM}}(\boldsymbol{Z}, \boldsymbol{\theta}^{\mathsf{SPO}}(\boldsymbol{Z}))$ and $\boldsymbol{\theta}^{\mathsf{SPO+}}(\boldsymbol{Z}))$, where $\boldsymbol{\theta}^{\mathsf{SPO}}(\boldsymbol{Z})$ and $\boldsymbol{\theta}^{\mathsf{SPO+}}(\boldsymbol{Z})$ are obtained by minimizing the so-called SPO and SPO+ losses, respectively. Other methods, for example, Wilder et al. (2019a), can be rewritten similarly. As mentioned, our analysis will bound the error when debiasing these polices as well.

Of course, we are not limited to a linear model for $r_j(z, \boldsymbol{\theta})$. We could alternatively use a nonlinear specification $r_j(z, \boldsymbol{\theta}) = f(\boldsymbol{W}_j, \boldsymbol{\theta})$ for some, given, nonlinear regression $f$ with parameters $\boldsymbol{\theta}$. This specification of $\boldsymbol{r}(\boldsymbol{Z}, \boldsymbol{\theta})$ *still* gives rise to a class of affine plug-in policies. Again, many policies in the literature, including estimate-then-optimize policies and SPO+ over nonlinear hypothesis classes, can be written in the form $\boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}(\boldsymbol{Z}))$ for some particular mapping $\boldsymbol{\theta}(\boldsymbol{Z}) \in \Theta$.

• **Mixed-Effects Policies:** When $\boldsymbol{W}_j$ is not informative for $\mu_j$, plug-ins for regression models perform poorly because no choice of $\boldsymbol{\theta}$ yields a good estimate of $\boldsymbol{\mu}$. By contrast, if $\nu_{\min}$ is large, SAA performs quite well. Mixed-effects policies interpolate between these choices. Define

$$\boldsymbol{x}^{\mathsf{ME}}(\boldsymbol{Z}, (\tau, \boldsymbol{\beta})) \in \arg\min_{x \in \mathcal{X}} \sum_{j=1}^{n} \left( \frac{\nu_j}{\nu_j + \tau} Z_j + \frac{\tau}{\nu_j + \tau} \boldsymbol{W}_j^\top \boldsymbol{\beta} \right) x_j, \tag{2.4}$$

where we have focused on a linear model for simplicity and made the dependence on $\boldsymbol{\theta} = (\tau, \boldsymbol{\beta})$ explicit for clarity. Mixed-effects policies are strongly motivated by Bayesian analysis (Gelman et al. 2014). These policies generalize the Bayes-inspired policy class considered in Gupta and Rusmevichientong (2021). Again, we observe that $\boldsymbol{x}^{\mathsf{ME}}(\boldsymbol{Z}, (\tau, \boldsymbol{\beta}))$ is an affine-plug in policy. Moreover,

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we can also consider shrinking toward a nonlinear regression model as in (Ignatiadis and Wager 2019).

In Definition 2.6, we require that $r_j(Z_j)$ depends only on $Z_j$ and not on $Z_k$ for $k \neq j$. We exploit this structure in the design and analysis of our debiasing technique. However, this requirement precludes certain types of plug-ins, for example, those based on linear smoothers (Buja et al. 1989), including local polynomial regression and k-nearest neighbors. Extension of our method to these settings remains an interesting open research question.

The choice of which affine plug-in policy class to use is largely application dependent. Our bounds in Section 4 provide some preliminary guidance, suggesting a tradeoff between the expressiveness of the policy class and the error of our estimator.

## 3. Variance Gradient Correction

We make the following assumption on problem parameters for the remainder of the paper.

**Assumption 3.1** (Assumptions on Parameters). *There exists a constant $C_\mu > 1$ such that $\|\boldsymbol{\mu}\|_\infty \leq C_\mu$, and constants $0 < \nu_{\min} < 1 < \nu_{\max} < \infty$ such that $\nu_{\min} \leq \nu_j \leq \nu_{\max}$ for all $j$. Moreover, we assume that $n \geq 3$.*

The assumptions for $C_\mu$ and $\nu_{\min}, \nu_{\max}$ are without loss of generality. These assumptions and the assumption on $n$ allow us to simplify the presentation of some results by absorbing lower order terms into leading constants.

The in-sample performance of a policy $\boldsymbol{x}(\boldsymbol{Z})$ is $\boldsymbol{Z}^\top \boldsymbol{x}(\boldsymbol{Z})$. Let $\boldsymbol{\xi} = \boldsymbol{Z} - \boldsymbol{\mu}$. We call the difference between in-sample and out-of-sample performance, corresponding to $(\boldsymbol{Z} - \boldsymbol{\mu})^\top \boldsymbol{x}(\boldsymbol{Z}) = \boldsymbol{\xi}^\top \boldsymbol{x}(\boldsymbol{Z})$, the *in-sample optimism*. The expected in-sample optimism $\mathbb{E}[\boldsymbol{\xi}^\top \boldsymbol{x}(\boldsymbol{Z})]$ is the in-sample bias.

Our method estimates the in-sample optimism of an affine, plug-in policy $\boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta})$. To this end, denote the plug-in objective value by

$$V(\boldsymbol{Z}, \boldsymbol{\theta}) \equiv r(\boldsymbol{Z}, \boldsymbol{\theta})^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}) = \min_{\boldsymbol{x} \in \mathcal{X}} r(\boldsymbol{Z}, \boldsymbol{\theta})^\top \boldsymbol{x}. \tag{3.1}$$

Because it is the minimum of linear functions, $\boldsymbol{Z} \longmapsto V(\boldsymbol{Z}, \boldsymbol{\theta})$ is always concave. We then estimate the in-sample optimism by the VGC defined by

$$D(\boldsymbol{Z}, (\boldsymbol{\theta}, h)) \equiv \sum_{j=1}^{n} D_j(\boldsymbol{Z}, (\boldsymbol{\theta}, h)), \tag{3.2}$$

where for $j = 1, 2, \ldots, n$,

$$D_j(\boldsymbol{Z}, (\boldsymbol{\theta}, h)) \equiv \begin{cases} \mathbb{E}\left[\dfrac{1}{h\sqrt{\nu_j} a_j(\boldsymbol{\theta})}(V(\boldsymbol{Z} + \delta_j \boldsymbol{e}_j) - V(\boldsymbol{Z})) \,\middle|\, \boldsymbol{Z}\right], & \text{if } a_j \neq 0, \\ 0, & \text{otherwise}, \end{cases} \tag{3.3}$$

and $\delta_1, \ldots, \delta_n$ are independent Gaussian random variables such that $\delta_j \sim \mathcal{N}\left(0, h^2 + \frac{2h}{\sqrt{\nu_j}}\right)$ for all $j$. To reduce notation, we define $\bar{\Theta} \equiv \Theta \times [h_{\min}, h_{\max}]$ for $0 < h_{\min} \leq h_{\max}$ and write $D(\boldsymbol{Z}, \boldsymbol{\theta})$ with $\boldsymbol{\theta} \in \bar{\Theta}$. We use the defined notation for results that require separating $h$ and $\boldsymbol{\theta}$.

The VGC is defined as a (conditional) expectation over the auxiliary random variables $\delta_j$. In practice, we can approximate this expectation to arbitrary precision by simulating $\delta_j$ and averaging (see Online Appendix B.5 for more efficient implementations).

Given the VGC, we estimate the out-of-sample performance by

$$\boldsymbol{\mu}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}) \approx \boldsymbol{Z}^\top \boldsymbol{x}(\boldsymbol{\theta}) - D(\boldsymbol{Z}, \boldsymbol{\theta}). \tag{3.4}$$

In Section 3.1, we motivate the VGC. We then establish some of its key properties, namely that it is almost an unbiased estimator for the in-sample optimism, its variance is often vanishing as $n \to \infty$, and it is smooth in the policy class.

**Policy Classes That Do Not Depend on $\boldsymbol{Z}$**: Recall our plug-ins for linear regression models example from Section 2.2. From Example 2.3, we can see that $D(\boldsymbol{Z}, (\boldsymbol{\theta}, h)) = 0$ uniformly over the class. Said differently, the in-sample performance is already an unbiased estimator of out-of-sample performance. This happy coincidence occurs whenever the plug-in function $r_j(z, \boldsymbol{\theta})$ does not depend on $z$ for each $j$ and $\boldsymbol{\theta}$. That said, we stress that although in-sample performance is an unbiased estimator, it is not immediately clear what the variance of this estimator is. We discuss this further in Sections 3.3 and 4. Moreover, when $r_j(z, \boldsymbol{\theta})$ *does* depend on $z$, the VGC is typically nonzero.

### 3.1. Motivating the VGC

Throughout this section, $\boldsymbol{\theta}$ is fixed so we drop it from the notation. Our heuristic derivation of $D(\boldsymbol{Z})$ proceeds in three steps.

Step 1**: Re-Expressing the In-Sample Optimism via Danskin's Theorem.** Fix some $j$. If $a_j = 0$, then from the plug-in policy problem (Problem (2.1)) we see that $\boldsymbol{x}(\boldsymbol{Z})$ is independent of $Z_j$ and the corresponding term in the in-sample bias is mean-zero, that is, $\mathbb{E}[\xi_j \boldsymbol{x}(\boldsymbol{Z})] = 0 = D_j(\boldsymbol{Z})$. In other words, we do not correct such terms. When $a_j \neq 0$, consider the function

$$\lambda \longmapsto V(\boldsymbol{Z} + \lambda \xi_j \boldsymbol{e}_j). \tag{3.5}$$

This function is an example of a parametric optimization problem. Danskin's theorem (Bertsekas 1997, section B.5) characterizes its derivative with respect to $\lambda$.[4] Specifically, for any $\lambda \in \mathbb{R}$ such that $\boldsymbol{x}(\boldsymbol{Z} + \lambda \xi_j \boldsymbol{e}_j)$ is the unique optimizer to Problem (2.1), we have

$$\frac{\partial}{\partial \lambda} V(\boldsymbol{Z} + \lambda \xi_j \boldsymbol{e}_j) = a_j \xi_j x_j(\boldsymbol{Z} + \lambda \xi_j \boldsymbol{e}_j).$$

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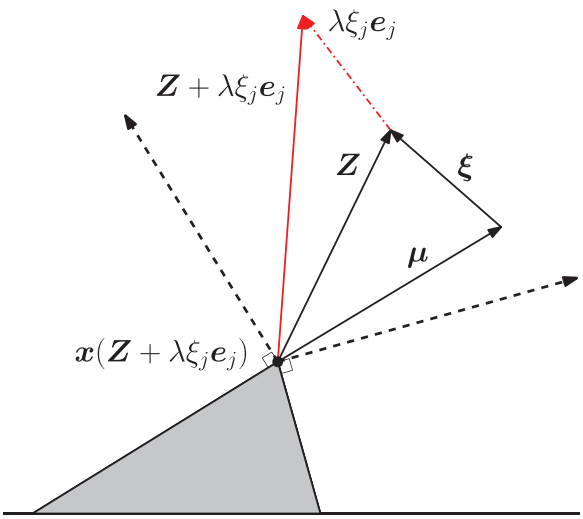
**Figure 2.** (Color online) Geometric Intuition of VGC




*Notes.* When $\mathcal{X}$ is polyhedral, $\boldsymbol{x}_j(\boldsymbol{Z}+\lambda\xi_j e_j)$ must occur at a vertex if it is unique. Hence, small perturbations to $\lambda$ do not change the solution, and the derivative of $V(\boldsymbol{Z}+\lambda\xi_j e_j)$ is entirely determined by the derivative of $\boldsymbol{r}(\boldsymbol{Z}+\lambda\xi_j e_j)$. Similar intuition holds for nonpolyhedral $\mathcal{X}$.

When $\boldsymbol{x}(\boldsymbol{Z}+\lambda\xi_j\boldsymbol{e}_j)$ is not the unique optimizer, $a_j\xi_j x_j(\boldsymbol{Z}+\lambda\xi_j\boldsymbol{e}_j)$ is a subgradient (see Figure 2 for intuition).

Notice that $\frac{\partial}{\partial\lambda}V(\boldsymbol{Z}+\lambda\xi_j\boldsymbol{e}_j)$ is the derivative of the plug-in value when we make the $j$th component of $\boldsymbol{Z}$ more variable, that is, variance increases by a factor $(1+\partial\lambda)^2$, where $\partial\lambda$ represents an infinitely small perturbation to $\lambda$. This observation motivates our nomenclature VGC.

Evaluating the previous derivative at $\lambda = 0$, dividing by $a_j$, and summing over $j$ such that $a_j \neq 0$ allows us to re-express the in-sample bias whenever $\boldsymbol{x}(\boldsymbol{Z})$ is unique as

$$\sum_{j=1}^{n}\xi_j x_j(\boldsymbol{Z}) = \sum_{j:a_j\neq 0}\frac{1}{a_j}\cdot\frac{\partial}{\partial\lambda}V(\boldsymbol{Z}+\lambda\xi_j\boldsymbol{e}_j)\Big|_{\lambda=0}.$$

Unfortunately, it is not clear how to evaluate these derivatives from the data. This leads to the second step in our derivation.

Step 2**: Approximating the Derivative via Randomized Finite Differencing.** As a first attempt, we approximate the above derivatives with first-order, forward finite-differences (LeVeque 2007, chapter 1). Intuitively, we expect that for a sufficiently small step size $h > 0$,

$$\frac{\partial}{\partial\lambda}V(\boldsymbol{Z}+\lambda\xi_j\boldsymbol{e}_j)\Big|_{\lambda=0} = \frac{1}{h\sqrt{v_j}}(V(\boldsymbol{Z}+h\sqrt{v_j}\xi_j\boldsymbol{e}_j)-V(\boldsymbol{Z})) + o_p(1) \text{ as } h\to 0, \tag{3.6}$$

which suggests that

$$\sum_{j=1}^{n}\xi_j x_j(\boldsymbol{Z}) = \sum_{j:a_j\neq 0}\left[\frac{1}{h\sqrt{v_j}a_j}(V(\boldsymbol{Z}+h\sqrt{v_j}\xi_j\boldsymbol{e}_j)-V(\boldsymbol{Z}))\right] + o_p(n) \text{ as } h\to 0. \tag{3.7}$$

Unfortunately, the right side of Equation (3.7) is not computable from the data, because we do not observe $\mu_j$ and hence do not observe $\xi_j = Z_j - \mu_j$.

To circumvent this challenge, recall that $\xi_j$ is Gaussian and independent across $j$, and let $\delta_j$ be the independent Gaussian random variables defined in the definition of the VGC (Equation (3.3)). A direct computation shows $\boldsymbol{Z}+h\sqrt{v_j}\xi_j\boldsymbol{e}_j \sim_d \boldsymbol{Z}+\delta_j\boldsymbol{e}_j$, because both $Z+h\sqrt{v_j}\xi_j\boldsymbol{e}_j$ and $Z_j+\delta_j\boldsymbol{e}_j$ are Gaussians with matching mean and covariances.[5] Hence, $V(\boldsymbol{Z}+h\sqrt{v_j}\xi_j\boldsymbol{e}_j)\sim_d V(\boldsymbol{Z}+\delta_j\boldsymbol{e}_j)$.

Inspired by this relation, we replace the unknown $V(\boldsymbol{Z}+h\sqrt{v_j}\xi_j\boldsymbol{e}_j)$ by $V(\boldsymbol{Z}+\delta_j\boldsymbol{e}_j)$ in our first-order, finite difference approximation, yielding a *randomized* finite difference:

$$\underbrace{\sum_{j=1}^{n}\xi_j x_j(\boldsymbol{\xi}+\boldsymbol{\mu};\boldsymbol{\theta})}_{\text{In-Sample Optimism}} \approx \underbrace{\sum_{j:a_j\neq 0}\left[\frac{1}{h\sqrt{v_j}a_j(\boldsymbol{\theta})}(V(\boldsymbol{\mu}+\boldsymbol{\xi}+\delta_j\boldsymbol{e}_j)-V(\boldsymbol{\mu}+\boldsymbol{\xi}))\right]}_{\text{Randomized Finite Difference}}. \tag{3.8}$$

Step 3**: De-Randomizing the Correction.** Finally, in the spirit of Rao-Blackwellization, we then de-randomize this correction by taking conditional expectations over $\boldsymbol{\delta}$. This de-randomization reduces the variability of our estimator and yields the VGC (Equation (3.3)).

**3.1.1. Higher-Order Finite Difference Approximations.** Our previous heuristic motivation uses a first-order finite difference approximation, and our later theoretical analysis focuses on this setting for simplicity. However, it is possible to use higher-order approximations, which in turn reduce the bias. Theoretical analysis of such higher-order approximations is tedious but not substantively different from the first-order case. Hence, it is omitted. In our experiments, we use a particular second-order approximation described in Online Appendix B.5.

## 3.2. Bias of Variance Gradient Correction

Our first main result shows that one can make the heuristic derivation of the previous section rigorous when quantifying the bias of the VGC.

**Theorem 3.2** (Bias of the Variance Gradient Correction). *Under Assumptions 2.4 and 3.1, for any $0 < h < 1/e$ and any affine, plug-in policy $\boldsymbol{x}(\boldsymbol{Z},\boldsymbol{\theta})$, there exists a constant $C$*

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*(depending on $v_{\min}$) such that*

$$0 \le \mathbb{E}\left[\sum_{j=1}^{n} \xi_j x_j(\boldsymbol{\mu}+\boldsymbol{\xi},\boldsymbol{\theta}) - \sum_{j=1}^{n} D_j(\boldsymbol{\mu}+\boldsymbol{\xi},\boldsymbol{\theta})\right] \le C \cdot hn \log\left(\frac{1}{h}\right).$$

Recall we expect that, in typical instances, the full-information performance of Problem (1.1) is $O(n)$ as $n \to \infty$. Thus, this theorem asserts that as long as $h$ is small, say $h = h_n = o(1)$ as $n \to \infty$, the bias of VGC is negligible relative to the true out-of-sample performance. In this sense, VGC is asymptotically unbiased for large $n$.

The proof of the theorem is in Online Appendix B.1 and proceeds similarly to our heuristic derivation but uses the following monotonicity property to precisely quantify the "little oh" terms.

**Lemma 3.3** (Monotonicity of Affine Plug-in Policies). *For any $\boldsymbol{z}$ and $j$, the function $t \longmapsto \boldsymbol{x}(\boldsymbol{z}+t_j)$ is nonincreasing if $a_j \ge 0$, and the function is nondecreasing if $a_j < 0$.*

Intuitively, the lemma holds because $\boldsymbol{z} \longmapsto V(\boldsymbol{z})$ is a concave function (it is the minimum of affine functions). In particular, $t \longmapsto V(\boldsymbol{z}+t)$ is also concave, and by Danskin's theorem, $d/dt\, V((\boldsymbol{z}+t)) = a_j x_j(\boldsymbol{z}+t)$, whenever $\boldsymbol{x}(\boldsymbol{z}+t)$ is unique. Informally, the lemma then follows because the derivative of a concave function is nonincreasing. Online Appendix B.1 provides a formal proof accounting for points of nondifferentiability.

Before proceeding, we remark that Theorem 3.2 holds with small modifications under mild violations of the independent Gaussian assumption (Assumption 2.4). Specifically, in cases where $v_j$ are not known but are estimated, the bias of the VGC constructed with the estimated $v_j$ increases by a small term depending on the accuracy of the precisions. See Lemma B.3 in the online appendix for formal statements and proof.

### 3.3. Variance of the VGC

As mentioned in the contributions, the parameter $h$ controls the tradeoffs between bias and variance in our estimator. Unfortunately, although Theorem 3.2 gives a direct analysis of the bias under mild assumptions, a precise analysis of the variance (or tail behavior) of the VGC is more delicate. In this section, we provide a loose, but intuitive bound on the variance of VGC that illustrates the types of problems for which our estimator should perform well. The main message of this section is that the VGC concentrates at its expectation so long as the policy $\boldsymbol{x}(\boldsymbol{Z})$ is "stable" in the sense that perturbing one element of $\boldsymbol{Z}$ does not cause $\boldsymbol{x}(\boldsymbol{Z})$ solution to change too much.

The main challenge in showing $D(\boldsymbol{Z})$ concentrates at its expectation is that $D(\boldsymbol{Z})$ is a sum of *dependent* random variables $D_j(\boldsymbol{Z})$. Worse, this dependence subtly hinges on the structure of Problem (1.1) and the plug-in policy problem (Problem (2.1)) and hence is not amenable to techniques based on mixing or bounding the correlations between terms. We require a different approach.

As a first step toward analyzing $D(\boldsymbol{Z})$, we upper bound its variance by a related, fully randomized estimator.

**Lemma 3.4** (Fully Randomized VGC). *Suppose that the solution $\boldsymbol{x}(\boldsymbol{Z})$ to Problem (2.1) is almost surely unique. For each $j$ such that $a_j \neq 0$, let*

$$D_j^{\mathsf{R}}(\boldsymbol{Z}) \equiv \frac{\delta_j}{h\sqrt{v_j}a_j} x_j(\boldsymbol{Z}+\delta_j \tilde{U}_j \boldsymbol{e}_j), \tag{3.9}$$

*where $\tilde{U}_j \sim \mathit{Uniform}[0,1]$ and $\delta_j$ is defined in Equation (3.3). Let $D^{\mathsf{R}}(\boldsymbol{Z}) = \sum_{j:a_j\neq 0} D_j^{\mathsf{R}}(\boldsymbol{Z})$ denote the fully randomized VGC. Then, for any $j$ such that $a_j \neq 0$,*

$$D_j(\boldsymbol{Z}) = \mathbb{E}\left[D_j^{\mathsf{R}}(\boldsymbol{Z}) \,|\, Z\right] \quad \textit{and} \quad \mathbb{V}\mathrm{ar}(\mathrm{D}(\boldsymbol{Z})) \le \mathbb{V}\mathrm{ar}(\mathrm{D}^{\mathsf{R}}(\boldsymbol{Z})).$$

**Proof.** Again, by Danskin's theorem and the fundamental theorem of calculus:

$$V(\boldsymbol{Z}+\delta_j\boldsymbol{e}_j) - V(\boldsymbol{Z}) = \int_0^{\delta_j} a_j x_j(\boldsymbol{Z}+t_j)dt = \int_0^1 a_j\delta_j \boldsymbol{x}_j(\boldsymbol{Z}+t\delta_j\boldsymbol{e}_j)dt$$
$$= \mathbb{E}\left[a_j\delta_j \boldsymbol{x}(\boldsymbol{Z}+\delta_j\tilde{U}_j) \,|\, \boldsymbol{Z}, \delta_j\right].$$

Scaling both sides by $1/(a_j h\sqrt{v_j})$ and taking expectations over $\delta_j$ proves the first statement. The second then follows from Jensen's inequality. □

We next propose upper bounding $\mathbb{V}\mathrm{ar}(D^{\mathsf{R}}(\boldsymbol{Z}))$ with the Efron-Stein inequality. In particular, let $\overline{\boldsymbol{Z}}$, $\overline{\boldsymbol{\delta}}$ and $\overline{\boldsymbol{U}}$ be i.i.d. copies of $\boldsymbol{Z}$, $\boldsymbol{\delta}$ and $\tilde{\boldsymbol{U}}$, respectively, and let $\boldsymbol{Z}^k$ denote the vector $\boldsymbol{Z}$ but with the $k$th component replaced by $\overline{Z}_k$. Define $\boldsymbol{\delta}^k$ and $\tilde{\boldsymbol{U}}^k$ similarly. Let $D^{\mathsf{R}}(\boldsymbol{Z},\boldsymbol{\delta},\tilde{\boldsymbol{U}})$ be the fully randomized VGC with dependence on all constituent random variables made explicit. Then, by the Efron-Stein inequality:

$$\mathbb{V}\mathrm{ar}(D^{\mathsf{R}}(\boldsymbol{Z})) \le \frac{1}{2}\sum_{k=1}^{n}\mathbb{E}\left[(D^{\mathsf{R}}(\boldsymbol{Z},\boldsymbol{\delta},\tilde{\boldsymbol{U}}) - D^{\mathsf{R}}(\boldsymbol{Z}^k,\boldsymbol{\delta},\tilde{\boldsymbol{U}}))^2\right] \tag{3.10a}$$
$$+\frac{1}{2}\sum_{k=1}^{n}\mathbb{E}\left[(D^{\mathsf{R}}(\boldsymbol{Z},\boldsymbol{\delta},\tilde{\boldsymbol{U}}) - D^{\mathsf{R}}(\boldsymbol{Z},\boldsymbol{\delta}^k,\tilde{\boldsymbol{U}}))^2\right] \tag{3.10b}$$
$$+\frac{1}{2}\sum_{k=1}^{n}\mathbb{E}\left[(D^{\mathsf{R}}(\boldsymbol{Z},\boldsymbol{\delta},\tilde{\boldsymbol{U}}) - D^{\mathsf{R}}(\boldsymbol{Z},\boldsymbol{\delta},\tilde{\boldsymbol{U}}^k))^2\right]. \tag{3.10c}$$

Recall that in the typical case, $\boldsymbol{\mu}^\top \boldsymbol{x}(\boldsymbol{Z}) = O_p(n)$. Hence, in what follows, we will focus on developing conditions

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for which the upper bound in Equations (3.10a) to (3.10c) is $o(n^2)$. Indeed, such a bound would suggest $D(\boldsymbol{Z}) - \mathbb{E}[D(\boldsymbol{Z})] = o_p(n)$; that is, the stochastic fluctuations in the VGC are negligible relative to the magnitude of the out-of-sample error for $n$ sufficiently large. With this perspective, it is not difficult to argue that both Equations (3.10b) and (3.10c) contribute at most $O\left(\frac{n}{h}\right)$ (see proof of Theorem 3.5). Thus, we focus on Equation (3.10a).

Consider the $k$ th element of the sum. Write

$$\begin{aligned}
&\left|D^{\mathsf{R}}(\boldsymbol{Z}) - D^{\mathsf{R}}(\boldsymbol{Z}^k)\right| \\
&\le \frac{1}{h a_k \nu_{\min}} \left| \sum_{j=1}^{n} \delta_j (x_j(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j) - x_j(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j)) \right| \\
&\le \frac{\|\boldsymbol{\delta}\|_2 \sqrt{n}}{h a_k \nu_{\min}} \cdot \left( \frac{1}{n} \sum_{j=1}^{n} \left| x_j(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j) - x_j(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j) \right|^2 \right)^{1/2},
\end{aligned} \tag{3.11}$$

where the last inequality follows from Cauchy-Schwarz's inequality. Because each $\delta_j$ is Gaussian, we expect $\|\boldsymbol{\delta}\|_2^2$ to concentrate sharply at its mean, that is, $\|\boldsymbol{\delta}\|_2^2 = O_p(hn)$. Thus, by squaring Equation (3.11), taking expectations, and substituting into Equation (3.10a), we roughly expect

$$\begin{aligned}
&\mathbb{V}\mathrm{ar}(D^{\mathsf{R}}(\boldsymbol{Z})) \\
&\le \underbrace{O\left(\frac{n}{h}\right)}_{\text{Problem (3.10b) and (3.10c)}} + \mathbb{E}\left[ \frac{\|\boldsymbol{\delta}\|_2^2 n^2}{h^2 \nu_{\min}^2 a_{\min^2}} \right. \\
&\qquad \left. \cdot \frac{1}{n^2} \sum_{k=1}^{n} \sum_{j=1}^{n} |x_j(\boldsymbol{Z} + \delta_j \tilde{U}_j) - x_j(\boldsymbol{Z}^k + \delta_j \tilde{U}_j)|^2 \right] \\
&\approx O\left(\frac{n}{h}\right) + O\left(\frac{n^3}{h}\right) \\
&\qquad \underbrace{\cdot \frac{1}{n^2} \sum_{k=1}^{n} \sum_{j=1}^{n} \mathbb{E}\left[ |x_j(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j) - x_j(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j)|^2 \right]}_{\text{Avg. Solution Instability}},
\end{aligned}$$

where $a_{\min} \equiv \min_{j: a_j \neq 0} |a_j|$.

We call the indicated term the average solution instability. In the worst case, it is at most one because $\mathcal{X} \subseteq [0,1]^n$. If, however, it were $O(n^{-\alpha})$ for some $\alpha > 1$, then the $\mathbb{V}\mathrm{ar}(D^{\mathsf{R}}(\boldsymbol{Z})) = o(n^2)$ as desired.

How do we intuitively interpret average solution instability? Roughly, in the limit as $h \to 0$, we might expect that $\boldsymbol{x}(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j) \approx \boldsymbol{x}(\boldsymbol{Z})$ because $\delta_j = O_p(\sqrt{h})$. Then the average solution instability is essentially the expected change in the solution in a randomly chosen component $j$ when we replace the data for a randomly chosen component $k$ with an i.i.d. copy. This interpretation suggests average solution instability should be small as long as a small perturbation to the $k$th component does not change the entire solution vector $\boldsymbol{x}(\boldsymbol{Z})$ by a large amount, that is, if small perturbations lead to small, local changes in the solutions. Intuitively, many large-scale optimization problems exhibit such phenomenon (Gamarnik 2013), so we broadly expect the VGC to have low variance.

The previous heuristic argument can be made formal as in the following theorem.

**Theorem 3.5** (Variance of the VGC). *Suppose that the solution $\boldsymbol{x}(\boldsymbol{Z})$ to Problem (2.1) is almost surely unique, that there exists a constant $C_1$ (not depending on n) such that $\mathbb{E}[\frac{1}{n^2} \sum_{k=1}^{n} \sum_{j=1}^{n} (x_j(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j) - x_j(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j))^2] \le C_1 n^{-\alpha}$ and that Assumption 3.1 holds. Then, there exists a constant $C_2$ (depending on $\nu_{\min}$ and $a_{\min} \equiv \min_j a_j$) such that for any $0 < h < 1/e$*

$$\mathbb{V}\mathrm{ar}(D(\boldsymbol{Z})) = \frac{C_2}{h} \max(n^{3-\alpha}, n).$$

*In particular, in the typical case where the full-information solution to Problem (1.1) is O(n), the stochastic fluctuations in the VGC are negligible relative to the out-of-sample performance if $\alpha > 1$.*

The proof of Theorem 3.5 is in Online Appendix B.4.

We remark that Theorem 3.5 provides a *sufficient* condition for the variance of the VGC to be negligible asymptotically and to show that $h$ controls the bias-variance tradeoff; however, the bound is not tight. In Section 4, we provide a tighter analysis given more stringent assumptions on Problems (1.1) and (1.2), which then also provides us guidance on how to select $h$ to approximately balance the bias-variance tradeoff.

### 3.4. Smoothness and Boundedness of the VGC

One of the key advantages of our VGC is that it is smooth in the policy class, provided $\boldsymbol{\theta} \longmapsto r(\cdot, \boldsymbol{\theta})$ is "well behaved." Other corrections, like the Stein correction of Gupta and Rusmevichientong (2021), do not enjoy such smoothness. In Section 5, we argue this smoothness improves the empirical performance of our method. We formalize well behaved in the next assumption.

**Assumption 3.6** (Plug-in Function Is Smooth). *We assume the functions $a_j(\boldsymbol{\theta})$, $b_j(\boldsymbol{\theta})$ are each L-Lipschitz continuous for all $j = 1, \ldots, n$. Moreover, we assume there exists $a_{\max}, b_{\max} < \infty$ such that*

$$\sup_{\boldsymbol{\theta} \in \Theta} |a_j(\boldsymbol{\theta})| \le a_{\max} \quad \text{and} \quad \sup_{\boldsymbol{\theta} \in \Theta} |b_j(\boldsymbol{\theta})| \le b_{\max} \quad \forall j = 1, \ldots, n.$$

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*Finally, we assume there exists* $a_{\min}$ *such that*

$$0 < a_{\min} \le \inf\{|a_j(\boldsymbol{\theta})| : a_j(\boldsymbol{\theta}) \neq 0, j = 1, \ldots, n, \boldsymbol{\theta} \in \Theta\}.$$

In words, Assumption 3.6 requires the functions $a_j(\boldsymbol{\theta})$ and $b_j(\boldsymbol{\theta})$ to be Lipschitz smooth, bounded, and that the nonzero components of $a_j(\boldsymbol{\theta})$ be bounded away from zero.

**3.4.1. Bias and Variance of VGC for Plug-in Linear Regression Models.** Recall our plug-in linear model class from Section 2.2. Because $a_j(\boldsymbol{\theta}) = 0$ for all $j$, $D(\boldsymbol{Z}, \boldsymbol{\theta}) = 0$ for all $\boldsymbol{Z}$ and (non–data-driven) $\boldsymbol{\theta}$ for this class. Said differently, the in-sample performance of a policy is, itself, our estimate of the out-of-sample performance, and both Theorems 3.2 and 3.5 can both be strengthened; the bias of our estimator and variance of the correction are both zero. More generally, $D(\boldsymbol{Z}, \boldsymbol{\theta}) = 0$ whenever the plug-in functionals $r_j(z, \boldsymbol{\theta})$ do not depend on $z$ for all $j$.

We stress, however, that this analysis does not immediately guarantee that the in-sample performance of policies of the form $\boldsymbol{x}^{\mathrm{LM}}(\boldsymbol{Z}, \boldsymbol{\theta}(\boldsymbol{Z}))$ is a good estimate of out-of-sample performance, because $\boldsymbol{\theta}(\boldsymbol{Z})$ depends on $\boldsymbol{Z}$. In Section 4, we provide sufficient conditions to ensure that in-sample performance is, indeed, a good estimate of out-of-sample performance. Moreover, when $\boldsymbol{r}(\boldsymbol{Z}, \boldsymbol{\theta})$ does depend on $\boldsymbol{Z}$, for example, as with our mixed effects regression class, $D(\boldsymbol{Z}, \boldsymbol{\theta})$ is generally nonzero.

**Lemma 3.7** (Smoothness of Variance Gradient Correction). *Under Assumptions 3.1 and 3.6, the following hold:*

(i) *There exists a constant* $C_1$ *(depending on* $a_{\min}, a_{\max}, b_{\max}$, *and* $\nu_{\min}$*) such that for any* $\boldsymbol{z} \in \mathbb{R}^n$ *and any* $0 < h < 1/e$, *the function* $\boldsymbol{\theta} \longmapsto D(z, (\boldsymbol{\theta}, h))$ *is Lipschitz continuous with parameter* $\frac{C_1 n^2 L}{h\sqrt{\nu_{\min}}}(\|\boldsymbol{z}\|_\infty + 1)$. *Moreover, there exists a constant* $C_2$ *(depending on* $C_\mu$ *and* $C_1$*) such that for any* $R > 1$, *with probability at least* $1 - e^{-R}$, *the (random) function* $\boldsymbol{\theta} \longmapsto D(\boldsymbol{Z}, (\boldsymbol{\theta}, h))$ *is Lipschitz continuous with parameter* $\frac{C_2 L}{h}\sqrt{\frac{R}{\nu_{\min}}} \cdot n^2\sqrt{\log n}$.

(ii) *Consider* $D(z, (\boldsymbol{\theta}, h))$, *where* $h \in [h_{\min}, h_{\max}]$ *and* $0 < h_{\max} - h_{\min} < 1$. *There exists an absolute constant* $C_3$ *such that for any* $\boldsymbol{z} \in \mathbb{R}^n$ *and* $\boldsymbol{\theta} \in \Theta$, *the following holds:*

$$\left|D(z, (\boldsymbol{\theta}, h)) - D(z, (\boldsymbol{\theta}, \bar{h}))\right| \le \frac{C_3 n}{h_{\min}\nu_{\min}^{3/4}}\sqrt{|h - \bar{h}|}.$$

See Online Appendix B.2 for a proof. Intuitively, the result follows because $\boldsymbol{\theta} \longmapsto V(z, \boldsymbol{\theta})$ is Lipschitz by Danskin's theorem and $D(z, \boldsymbol{\theta})$ is a linear combination of such functions. The second part follows from a high-probability bound on $\|\boldsymbol{Z}\|_\infty$.

In addition to being smooth, the VGC is also bounded as a direct result of taking the conditional expectation over the perturbation parameters $\delta_j$.

**Lemma 3.8** (VGC Is Bounded). *Suppose that Assumptions 3.1 and 3.6 hold. For any* $\boldsymbol{z}$, *and any* $j = 1, \ldots, n$,

$$|D_j(\boldsymbol{z})| \le \frac{\sqrt{3}}{\nu_{\min}^{3/4}\sqrt{h}}.$$

The proof can be found in Online Appendix B.2. The result follows from observing that the $j$th component of the VGC is the difference of the optimal objectives values of two optimization problems whose cost vector differs by $O(|\delta_j|)$ in one component. Thus, the two optimal objective values can only differ by $O(|\delta_j|)$, which is at most a constant once we take the conditional expectation.

## 4. Estimating Out-of-Sample Performance for Weakly Coupled Problems

In this section, we provide high-probability tail bounds on the error of our estimator for out-of-sample performance that hold uniformly over a given policy class. Such bounds justify using estimator for policy learning, that is, identifying the best policy within the class. They are also substantively stronger than the variance analysis of Theorem 3.5 as they provide exponential bounds on the tail behavior, rather than bounding the second moment. Additionally, we show the uniform results hold even when $\boldsymbol{\theta} \in \bar{\Theta}$ is chosen in a data-driven manner (which recall also includes $h$).

From the definition of the VGC out-of-sample estimator (Equation (3.4)), the error of our estimator of out-of-sample performance for $\boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}(\boldsymbol{Z}))$ is

$$\underbrace{\left|\boldsymbol{\mu}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}(\boldsymbol{Z})) - (\boldsymbol{Z}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}(\boldsymbol{Z})) - D(\boldsymbol{Z}, \boldsymbol{\theta}(\boldsymbol{Z})))\right|}_{\text{Error Estimating Out of Sample Perf.}}$$

$$\le \sup_{\boldsymbol{\theta} \in \bar{\Theta}} \underbrace{\left|\boldsymbol{\xi}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}) - D(\boldsymbol{Z}, \boldsymbol{\theta})\right|}_{\text{Error Estimating In-Sample Optimism}}$$

$$\le \sup_{\boldsymbol{\theta} \in \bar{\Theta}} \left|\boldsymbol{\xi}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}) - \mathbb{E}\left[\boldsymbol{\xi}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta})\right]\right| \tag{4.1a}$$

$$+ \sup_{\boldsymbol{\theta} \in \bar{\Theta}} |D(\boldsymbol{Z}, \boldsymbol{\theta}) - \mathbb{E}[D(\boldsymbol{Z}, \boldsymbol{\theta})]| \tag{4.1b}$$

$$+ \sup_{\boldsymbol{\theta} \in \bar{\Theta}} \left|\mathbb{E}\left[\boldsymbol{\mu}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}) - \boldsymbol{Z}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}) - D(\boldsymbol{Z}, \boldsymbol{\theta})\right]\right|. \tag{4.1c}$$

We bounded Equation (4.1c) in Theorem 3.2. Our goal will be to find sufficient conditions to show the remaining terms are also $o_p(n)$ uniformly over $\boldsymbol{\theta} \in \Theta$. Then, in the typical case where the out-of-sample performance is $O_p(n)$, the error of our estimator will be negligible relative to the true out-of-sample performance. Our strategy will be to leverage empirical process theory because the argument of each suprema is a sum of

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random variables. Importantly, this empirical process analysis does *not* strictly require the independent Gaussian assumption (Assumption 2.4). The challenge of course is that the constraints of Problem (2.1) introduce a complicated dependence between the terms.

Inspired by the average stability condition of Theorem 3.5, we focus on classes of "weakly coupled" optimization problems. We consider two such classes of problems, those weakly coupled by variables in Section 4.1 and those weakly coupled by constraints in Section 4.2. We provide formal definitions here.

### 4.1. Problems Weakly Coupled by Variables

We say an instance of Problem (1.1) is weakly coupled by variables if fixing a small number of variables causes the problem to separate into many, decoupled subproblems. Generically, such problems can be written as

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & (\boldsymbol{\mu}^0)^\top \boldsymbol{x}^0 + \sum_{k=1}^{K} \left(\boldsymbol{\mu}^k\right)^\top \boldsymbol{x}^k \\ \text{s.t.} \quad & \boldsymbol{x}^0 \in \mathcal{X}^0, \quad \boldsymbol{x}^k \in \mathcal{X}^k(\boldsymbol{x}^0), \quad \forall k = 1, \ldots, K. \end{aligned} \tag{4.2}$$

Here, $\boldsymbol{x}^0$ represents the coupling variables and $k = 1, \ldots, K$ represent distinct subproblems. Once $\boldsymbol{x}^0$ is fixed, each subproblem can be solved separately. Intuitively, if $\dim(\boldsymbol{x}^0)$ is small relative to $n$, the subproblems of Equation (4.2) are only "weakly" coupled. Some reflection shows both Examples 2.2 and 2.3 from Section 2 are weakly coupled by variables.

Let $S_k \subseteq \{1, \ldots, n\}$ be the indices corresponding to $\boldsymbol{x}^k$ for $k = 0, \ldots, K$, and $S_{\max} = \max_{k \geq 0} |S_k|$. The sets $S_0, \ldots, S_K$ form a disjoint partition of $\{1, \ldots, n\}$. Without loss of generality, reorder the indices so that the $S_k$ occur in order; that is, $(j : j \in S_0), \ldots, (j : j \in S_K)$ is a consecutive sequence.

Given the weakly coupled structure of Equation (4.2), we define a generalization of $\boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta})$: For each $\boldsymbol{x}^0 \in \mathcal{X}^0$ and $\boldsymbol{\theta} \in \bar{\Theta}$, let

$$\boldsymbol{x}^k(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0) \in \operatorname*{arg\,min}_{\boldsymbol{x}^k \in \mathcal{X}^k(\boldsymbol{x}^0)} \boldsymbol{r}^k(\boldsymbol{Z}, \boldsymbol{\theta})^\top \boldsymbol{x}^k, \quad k = 1, \ldots, K, \tag{4.3}$$

where $\boldsymbol{r}^k(\boldsymbol{Z}, \boldsymbol{\theta}) = (r_j(\boldsymbol{Z}, \boldsymbol{\theta}) : j \in S_k)$. Intuitively, the vector

$$\boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0) \equiv ((\boldsymbol{x}^0)^\top, \boldsymbol{x}^1(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0)^\top, \ldots, \boldsymbol{x}^K(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0)^\top)^\top$$

satisfies the average instability condition of Theorem 3.5 as long as $S_{\max}$ is not too large because the $j$th component of the solution changes when perturbing the $k$th data point if and only if $j$ and $k$ belong to the same subproblem. This event happens with probability at most $S_{\max}/n^2$.

The key to making this intuition formal and obtaining exponential tails for the error of the out-of-sample estimator (Equation 4.1)) is that

$$\begin{aligned} &\sup_{\boldsymbol{\theta} \in \bar{\Theta}} \left| \boldsymbol{\xi}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}) - \mathbb{E}\left[\boldsymbol{\xi}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta})\right] \right| \\ &\quad \leq \sup_{\boldsymbol{\theta} \in \bar{\Theta}, \boldsymbol{x}^0 \in \mathcal{X}^0} \left| \boldsymbol{\xi}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0) - \mathbb{E}\left[\boldsymbol{\xi}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0)\right] \right| \quad \text{and,} \\ &\sup_{\boldsymbol{\theta} \in \bar{\Theta}} \left| D(\boldsymbol{Z}, \boldsymbol{\theta}) - \mathbb{E}[D(\boldsymbol{Z}, \boldsymbol{\theta})] \right| \\ &\quad \leq \sup_{\boldsymbol{\theta} \in \bar{\Theta}, \boldsymbol{x}^0 \in \mathcal{X}^0} \left| D(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0) - \mathbb{E}\left[D(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0)\right] \right|, \end{aligned}$$

where both $\boldsymbol{\xi}^\top \boldsymbol{x}(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0)$ and $D(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0)$ can, for a fixed $\boldsymbol{\theta}, \boldsymbol{x}^0$, be seen as sums of $K$ *independent* random variables. To obtain uniform bounds, we then need to control only the metric entropy of the resulting (lifted) stochastic processes indexed by $(\boldsymbol{\theta}, \boldsymbol{x}^0)$.

We propose a simple assumption on the policy class to control this metric entropy. We believe this assumption is easier to verify than other assumptions used in the literature (e.g., bounded linear subgraph dimension or bounded Natarajan dimension) but admittedly slightly more stringent.

**Assumption 4.1** (Lifted Affine Plug-in Policy). *Given an affine plug-in policy class defined by $\boldsymbol{r}(\cdot, \boldsymbol{\theta})$ for $\boldsymbol{\theta} \in \Theta$, we say this class satisfies the lifted affine plug-in policy assumption for problems weakly coupled by variables (Equation (4.2)) if there exists mapping $\phi(\cdot)$ and mappings $g_k(\cdot)$ for $k = 1, \ldots, K$ such that*

$$\boldsymbol{x}^k(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0) \in \operatorname*{arg\,min}_{\boldsymbol{x}^k \in \mathcal{X}^k(\boldsymbol{x}^0)} \phi(\boldsymbol{\theta})^\top g_k(\boldsymbol{Z}^k, \boldsymbol{x}^k, \boldsymbol{x}^0) \quad k = 1, \ldots, K, \quad \forall \boldsymbol{x}^0 \in \mathcal{X}^0.$$

We stress that the mapping $\phi(\cdot)$ is common to all $K$ subproblems and all $\boldsymbol{x}^0 \in \mathcal{X}^0$, and both $\phi(\cdot)$ and $g_k(\cdot)$ can be arbitrarily nonlinear. Moreover, $g_k(\cdot)$ may implicitly depend on the precisions $\boldsymbol{\nu}$ and covariates $\boldsymbol{W}$ as these are fixed constants. With the exception of policies from linear smoothers, each of our examples from Section 2.2 satisfies Assumption 4.1. For example, for plug-ins for linear regression models, we can simply take $\phi(\boldsymbol{\theta}) = \boldsymbol{\theta}$ and $g_k(\boldsymbol{Z}^k, \boldsymbol{x}^k, \boldsymbol{x}^0) = \sum_{j \in S_k} \boldsymbol{W}_j^\top x_j$.

When $\boldsymbol{x}^k(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0)$ is not the unique minimizer to the problem weakly coupled variables defined in Equation (4.3), we require that ties are broken consistently. Let $\mathrm{Ext}(\mathrm{Conv}(\mathcal{X}^k(\boldsymbol{x}^0)))$ denote the set of extreme points of $\mathrm{Conv}(\mathcal{X}^k(\boldsymbol{x}^0))$ and let $\mathcal{X}_{\max} = \max_{k \geq 0} \mathrm{Ext}(\mathrm{Conv}(\mathcal{X}^k(\boldsymbol{x}^0)))$. If $\boldsymbol{x}^k(\boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{x}^0)$ is unique, it is an extreme point.

**Assumption 4.2** (Consistent Tie-Breaking). *We assume there exists functions $\sigma_{k\boldsymbol{x}^0} : 2^{\mathcal{X}^k(\boldsymbol{x}^0)} \longmapsto \mathrm{Ext}(\mathrm{Conv}(\mathcal{X}^k(\boldsymbol{x}^0)))$*

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*such that*

$$x^k(Z,\theta,x^0) = \sigma_{kx^0}\left(\arg\min_{x^k\in\mathcal{X}^k(x^0)} \phi(\theta)^\top g_k(Z^k,x^k,x^0)\right) \qquad k=1,\ldots,K, \quad \forall x^0\in\mathcal{X}^0.$$

Consistent tie-breaking requires that if $(\theta_1, x_1^0)$ and $(\theta_2, x_2^0)$ induce the same minimizers in Equation (4.3) for some $Z$, then $x^k(Z,\theta_1,x_1^0) = x^k(Z,\theta_2,x_2^0)$, and this point is an extreme point of $\mathcal{X}^k(x^0)$.

Assumptions 4.1 and 4.2 allow us to bound the cardinality of the set $\{(x^0, x^1(Z,\theta,x^0),\ldots,x^K(Z,\theta,x^0)) : x^0 \in \mathcal{X}^0, \theta\in\Theta\}$ by adapting a geometric argument counting regions in a hyperplane arrangements from Gupta and Kallus (2022, lemma C.7). The cardinality of the set characterizes the metric entropy of the policy class.

Finally, for this section, we say a constant $C$ is *dimension-independent* if $C$ does *not* depend on $\{K, S_{\max}, h, \mathcal{X}^0, \mathcal{X}_{\max}, \dim(\phi)\}$ but may depend on $\{\nu_{\min}, C_\mu, L\}$. We now present the main result of this section.

**Theorem 4.3** (Policy Learning for Problems Weakly Coupled by Variables). *Suppose Assumptions 2.4, 3.1, 3.6, 4.1, and 4.2 all hold. Let $\mathcal{X}_{\max} \geq |\operatorname{Ext}(\operatorname{Conv}(\mathcal{X}^k(x^0)))|$ for all $k=1,\ldots,K$ and $x^0\in\mathcal{X}^0$, and assume $\mathcal{X}_{\max}<\infty$. Then, for $0<h_{\min}\leq h_{\max}\leq 1$, there exists a dimension-independent constant $C$ such that, for any $R>1$, with probability at least $1-2\exp(-R)$,*

$$\begin{aligned}\sup_{\theta\in\bar{\Theta}} \left|\xi^\top x(Z,\theta) - D(Z,\theta,h)\right| &\leq CKS_{\max}\cdot h_{\max}\log\left(\frac{1}{h_{\min}}\right)\\ &+ CS_{\max}R\sqrt{\frac{K\log(1+|\mathcal{X}^0|)}{h_{\min}}}\\ &\left(\log(K)\sqrt{\log(S_{\max})\log\left(h_{\min}^{-1}\cdot N\left(\sqrt{\frac{h_{\min}}{Kn^2}},\Theta\right)\right)}\right.\\ &+ \left.\sqrt{\log(K)\ \dim(\phi)\log(1+\mathcal{X}_{\max})}\right).\end{aligned}$$

*Here, $N(\epsilon,\Theta)$ is the $\epsilon$-covering number of the set $\Theta$.*

In the typical case where $\Theta$ does not depend on $n$ or $K$ and

$$\max(S_{\max}, |\mathcal{X}_0|,\ \dim(\phi)) = \tilde{O}(1) \text{ as } K\to\infty, \tag{4.4}$$

we can approximately minimize the previous bound by selecting $h\equiv h_k = O(K^{-1/3})$ and noting the relevant covering number grows at most logarithmically in $K$. This choice of $h$ approximately balances the deterministic and stochastic contributions, and the bound reduces to $\tilde{O}(C_{\mathrm{PI}}K^{2/3})$ for some $C_{\mathrm{PI}}$ (depending on $|\mathcal{X}_0|$, $\dim(\phi)$, $\dim(\theta)$) that measures the complexity of the policy class. Many applications satisfy the conditions in Equation (4.4), including the drone-assisted emergency medical response application (Example 2.2).

To illustrate, recall Example 2.2. Here, $y$ represents the binding variables "$x^0$," and we see $|\mathcal{X}^0| = \binom{L}{B}$. Moreover, $S_{\max} = L$, because $x$ decouples across $k$. Inspecting the constraints, $\mathcal{X}_{\max}\leq B$, because for each $k$, we choose exactly one depot from which to serve location $k$, and there are most $B$ available depots. Finally, for a fixed policy class, $\dim(\phi)$ is constant and the log covering number grows at most logarithmically in $K$. Most importantly, we expect $L$ (the number of possible depots) and $B$ (the budget) to be fairly small relative to $K$ because regulations and infrastructure limit placement of depots, but there are many possible locations for cardiac events. Here typical instances of Example 2.2 satisfy Equation (4.4). We return to Example 2.2 in Section 5 where we study the performance of our method numerically.

### 4.2. Problems with Weakly Coupled by Constraints

An instance of Problem (1.1) is weakly coupled by constraints if, after removing a small number of binding constraints, the problem decouples into many separate subproblems. Data-driven linear optimization problems of this form have been well studied by Li and Ye (2022) and Gupta and Rusmevichientong 2021. To facilitate comparisons to the existing literature, we study the specific problem

$$\max_{x\in\mathcal{X}} \sum_{j=1}^n \mu_j x_j, \qquad \mathcal{X} = \left\{x\in[0,1]^n : \frac{1}{n}\sum_{j=1}^n A_j x_j \leq b\right\}, \tag{4.5}$$

and corresponding plug-in policies

$$x(Z,\theta) \in \arg\max_{x\in\mathcal{X}} \sum_{j=1}^n r_j(Z_j,\theta)x_j. \tag{4.6}$$

Here, $A_j\in\mathbb{R}^m$ with $m\geq 1$. In particular, we consider a maximization instead of a minimization.

We next introduce a dual representation of Problem (4.6). Specifically, scaling the objective of Problem (4.6) by $\frac{1}{n}$ and dualizing the binding constraints yields

$$\lambda(Z,\theta)\in\arg\min_{\lambda\in\mathbb{R}_+^m}\left\{b^\top\lambda + \max_{x\in[0,1]^n}\frac{1}{n}\sum_{j=1}^n (r_j(Z_j,\theta) - A_j^\top\lambda)x_j\right\}. \tag{4.7}$$

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For a fixed $\boldsymbol{\lambda}$, the inner maximization of Equation (4.7) can be solved explicitly, yielding

$$\boldsymbol{\lambda}(\boldsymbol{Z},\boldsymbol{\theta}) \in \arg\min_{\boldsymbol{\lambda}\in\mathbb{R}^m_+} \mathcal{L}(\boldsymbol{\lambda},\boldsymbol{Z},\boldsymbol{\theta}), \quad \text{where}$$

$$\mathcal{L}(\boldsymbol{\lambda},\boldsymbol{Z},\boldsymbol{\theta}) \equiv \boldsymbol{b}^\top\boldsymbol{\lambda} + \frac{1}{n}\sum_{j=1}^n \left(r_j(Z_j,\boldsymbol{\theta}) - \boldsymbol{A}_j^\top\boldsymbol{\lambda}\right)^+.$$

By strong duality, $V(\boldsymbol{Z},\boldsymbol{\theta}) = \boldsymbol{r}(\boldsymbol{Z},\boldsymbol{\theta})^\top \boldsymbol{x}(\boldsymbol{Z},\boldsymbol{\theta}) = n\mathcal{L}(\boldsymbol{\lambda}(\boldsymbol{Z},\boldsymbol{\theta}),\boldsymbol{Z},\boldsymbol{\theta})$.

This dual representation highlights the weakly coupled structure. Indeed, the dependence across terms in the sum in $\mathcal{L}(\boldsymbol{\lambda}(\boldsymbol{Z},\boldsymbol{\theta}),\boldsymbol{Z},\boldsymbol{\theta})$ arises because $\boldsymbol{\lambda}(\boldsymbol{Z},\boldsymbol{\theta})$ depends on the entire vector $\boldsymbol{Z}$. However, this dependence has to be "channeled" through the $m$-dimensional vector $\boldsymbol{\lambda}(\boldsymbol{Z},\boldsymbol{\theta})$, and hence, when $m$ is small relative to $n$, cannot create too much dependence between the summands. Indeed, we will show that if $m$ is small relative to $n$, then $\boldsymbol{\lambda}(\boldsymbol{Z},\boldsymbol{\theta})$ concentrates at its expectation, that is, a constant, as $n \to \infty$, and hence the summands become independent asymptotically. This insight is key to the analysis.

To formalize these ideas, we make assumptions similar to those in Gupta and Rusmevichientong (2021) and Li and Ye (2022).

**Assumption 4.4** ($s_0$-Strict Feasibility). *There exists an $s_0 > 0$ and $\boldsymbol{x}_0 \in \mathcal{X}$ such that $\frac{1}{n}\sum_{j=1}^n \boldsymbol{A}_j x_j^0 + s_0 \le \boldsymbol{b}$.*

**Assumption 4.5** (Regularity of Matrix $A$). *There exists a constant $C_A$ such that $\|\boldsymbol{A}_j\|_\infty \le C_A$ for all $1 \le j \le n$. Moreover, there exists a constant $\beta > 0$ such that the minimal eigenvalue of $\frac{1}{n}\sum_{j=1}^n \boldsymbol{A}_j\boldsymbol{A}_j^\top$ is at least $\beta$.*

The strict feasibility assumption can often be satisfied by perturbing $\boldsymbol{A}$ or $\boldsymbol{b}$ and ensures the dual optimal values $\boldsymbol{\lambda}(\boldsymbol{Z},\boldsymbol{\theta})$ are bounded with high probability. The regularity assumptions on $\boldsymbol{A}$ ensure the function $\boldsymbol{\lambda} \longmapsto \mathbb{E}[\mathcal{L}(\boldsymbol{\lambda},\boldsymbol{Z},\boldsymbol{\theta})]$ is strongly convex, a key property in our proof. Such a property holds, for example, if the columns $\boldsymbol{A}_j$ are drawn randomly from some distribution.

Like Section 4.1, to obtain uniform bounds, we must also control the metric entropy of the different stochastic error terms in out-of-sample error in Equation (4.1). Generalizing Gupta and Rusmevichientong (2021), we make the following assumption.

**Assumption 4.6** (VC Policy Class). *There exists a function $\rho(\cdot)$ such that*

$$r_j(z_j,\boldsymbol{\theta}) = \rho((z_j,\nu_j,\boldsymbol{W}_j,\boldsymbol{A}_j),\boldsymbol{\theta}), \quad j = 1,\ldots,n$$

*and a constant $V$ such that the class of functions*

$$\left\{(z,\nu,\boldsymbol{W},\boldsymbol{A}) \longmapsto \rho((z,\nu,\boldsymbol{W},\boldsymbol{A}),\boldsymbol{\theta}) - \boldsymbol{A}^\top\boldsymbol{\lambda} : \boldsymbol{\theta}\in\Theta, \boldsymbol{\lambda}\in\mathbb{R}^m_+\right\}$$

*has a pseudo-dimension at most $V$. Without loss of generality, we further assume $V \ge \max(m,2)$.*

The size of the constant $V$ captures the complexity of the policy class that typically depends on the dimension of $\boldsymbol{\theta}$ and the number of binding constraints $m$. As an illustration, recall the plug-in for linear regression models policy class from Section 2. By Pollard (1990, lemma 4.4), this policy class satisfies Assumption 4.6 with $V = \dim(\boldsymbol{\theta}) + m$.

Finally, we say a constant $C$ is *dimension-independent* if $C$ does *not* depend on $\{n,h,m,V,\dim(\boldsymbol{\theta})\}$ but may depend on $\{\nu_{\min}, C_A, C_\mu, \beta, s_0, a_{\min}, a_{\max}, b_{\max}, L\}$.

The main result of this section is then as follows.

**Theorem 4.7** (Estimation Error for Problems Weakly Coupled Constraints). *Under Assumptions 2.4, 3.1, 3.6, and 4.4 to 4.6, for $0 < h_{\min} \le h_{\max} \le 1$, there exists dimension-independent constants $C$ and $n_0$, such that for any $R > 1$ and all $n \ge n_0 e^R$, we have that with probability $1 - C\exp(-R)$:*

$$\sup_{\boldsymbol{\theta}\in\bar{\Theta}} \left|\sum_{j=1}^n \xi_j x_j(\boldsymbol{Z},\boldsymbol{\theta}) - D_j(\boldsymbol{Z},\boldsymbol{\theta})\right|$$
$$\le Cn\cdot h_{\max}\log\left(\frac{1}{h_{\min}}\right)$$
$$+ C\cdot V^3\log^3 V\cdot R\cdot\sqrt{n\log\left(n\cdot N(n^{-3/2},\Theta)\right)}\cdot\frac{\log^5 n}{h_{\min}}.$$

To build intuition, consider instances where $\Theta$ does not depend on $n$. Then, $V = O(1)$, and the covering number grows at most logarithmically as $n \to \infty$. We can then minimize the bound (up to logarithmic terms) by taking $h \equiv h_n = O(n^{-1/4})$, yielding a bound of order $\tilde{O}(n^{3/4})$. In particular, in the typical instance where the full-information optimum (Problem (4.5)) is $O(n)$, the relative error of our estimate is $\tilde{O}(n^{-1/4})$, which is vanishing as $n \to \infty$.

**Remark 4.8.** The rate earlier ($\tilde{O}(n^{-1/4})$) is slightly slower than the rate of convergence of the Stein correction in Gupta and Rusmevichientong (2021) ($\tilde{O}(n^{-1/3})$). We attribute this difference to our choice of a first-order finite difference when constructing the VGC. A heuristic argument strongly suggests that had we instead used a second-order forward finite difference scheme as in Online Appendix B.5, we would recover the rate $\tilde{O}(n^{-1/3})$. Moreover, our numerical experiments (with the second order scheme) in Section 5 shows our second-order VGC to be very competitive.

**4.2.1. Proof Intuition: Approximate Strong-Convexity and Dual Stability.** To build intuition, recall that to show the convergence of VGC, it suffices to bound the average solution instability defined in Theorem 3.5. By complementary slackness, $x_j(\boldsymbol{Z}) = \mathbb{I}\left\{r_j(Z_j) > \boldsymbol{A}_j^\top\boldsymbol{\lambda}(\boldsymbol{Z})\right\}$ except possibly for $m$ fractional components. Hence, by rounding the fractional components, we have, for

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$j \neq k$,

$$\begin{aligned} & x_j(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j) - x_j(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j) \\ & \quad \leq \mathbb{I}\Big\{ r_j(Z_j + \delta_j \tilde{U}_j \boldsymbol{e}_j) \geq \boldsymbol{A}_j^\top \boldsymbol{\lambda}(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j) \Big\} \\ & \qquad - \mathbb{I}\Big\{ r_j(Z_j + \delta_j \tilde{U}_j \boldsymbol{e}_j) > \boldsymbol{A}_j^\top \boldsymbol{\lambda}(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j) \Big\} \\ & \quad \leq \mathbb{I}\Big\{ r_j(Z_j + \delta_j \tilde{U}_j \boldsymbol{e}_j) \in \Big\langle \boldsymbol{A}_j^\top \boldsymbol{\lambda}(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j), \\ & \qquad \boldsymbol{A}_j^\top \boldsymbol{\lambda}(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j) \Big\rangle \Big\}, \end{aligned}$$

where we use $\langle l, u \rangle$ to denote the interval $[\min(l,u), \max(l,u)]$. By symmetry, the same bound holds for $x_j(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j) - x_j(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j)$. Because summands where $j = k$ each contribute at most one to the average solution instability, we thus have that

$$\begin{aligned} & \frac{1}{n^2} \sum_{k=1}^{n} \sum_{j=1}^{n} \Big( x_j(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j) - x_j(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j) \Big)^2 \\ & \quad \leq \frac{1}{n^2} \sum_{k=1}^{n} \sum_{j=1}^{n} \mathbb{I}\Big\{ r_j(Z_j + \delta_j \tilde{U}_j \boldsymbol{e}_j) \in \Big\langle \boldsymbol{A}_j^\top \boldsymbol{\lambda}(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j), \\ & \qquad \boldsymbol{A}_j^\top \boldsymbol{\lambda}(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j) \Big\rangle \Big\} + O_p\Big(\frac{1}{n}\Big). \end{aligned}$$

The principal driver of the solution instability is the first double sum; in the worst case, it might be $O_p(1)$. It will be small if $\|\boldsymbol{\lambda}(\boldsymbol{Z} + \delta_j \tilde{U}_j \boldsymbol{e}_j) - \boldsymbol{\lambda}(\boldsymbol{Z}^k + \delta_j \tilde{U}_j \boldsymbol{e}_j)\|$ is small for most $j$ and $k$. Said differently, problems like Problem (4.6) that are weakly coupled by constraints will have small solution instability if the dual solutions $\boldsymbol{\lambda}(\cdot)$ are themselves stable, that is, if the dual solution does not change very much when we perturb one data point. Our analysis thus focuses on establishing this dual solution stability.

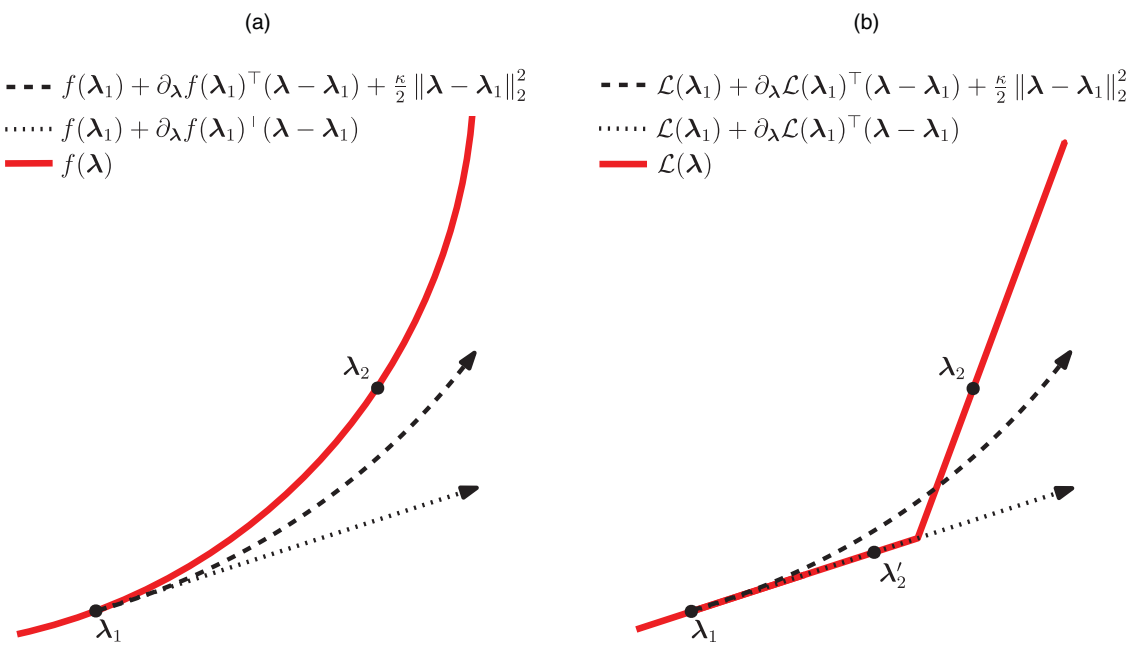
Unfortunately, solutions of linear optimization problems need not be stable: A small perturbation to the cost vector might cause the optimal solution to switch between extreme points, inducing a large change. By contrast, solutions of convex optimization problems with strongly convex objectives *are* stable (Shalev-Shwartz et al. 2010). The next key step of our proof is to show that, although $\boldsymbol{\lambda} \longmapsto \mathcal{L}(\boldsymbol{\lambda}, \boldsymbol{Z})$ is not strongly convex, it is still "approximately" strongly convex with high probability in a certain sense.

To be more precise, recall that a function $f(\boldsymbol{\lambda})$ is $\kappa$-strongly convex if

$$f(\boldsymbol{\lambda}_2) - f(\boldsymbol{\lambda}_1) \geq \nabla_{\boldsymbol{\lambda}} f(\boldsymbol{\lambda}_1)^\top (\boldsymbol{\lambda}_2 - \boldsymbol{\lambda}_1) + \frac{\kappa}{2} \|\boldsymbol{\lambda}_2 - \boldsymbol{\lambda}_1\|_2^2, \quad \forall \boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2 \in \mathrm{Dom}(f). \tag{4.8}$$

Here, $\kappa > 0$ and $\nabla_{\boldsymbol{\lambda}}$ denotes the subgradient. The left panel of Figure 3 depicts this condition graphically. For any two points $\boldsymbol{\lambda}_2$ and $\boldsymbol{\lambda}_1$, the first-order Taylor series underestimates the function value, and one can

**Figure 3.** (Color online) Approximate Strong Convexity of $\mathcal{L}(\boldsymbol{\lambda})$




*Notes.* (a) Strongly convex function $f(\boldsymbol{\lambda})$ and strong convexity condition Equation (4.8). (b) Because $\mathcal{L}(\boldsymbol{\lambda})$ is piecewise linear, it does not satisfy Equation (4.8) for points on same line segment ($\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2'$). However, when $\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2$ are sufficiently far apart, they are on different line segments, and Equation (4.8) is satisfied.

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"squeeze in" a quadratic correction $\frac{\kappa}{2}\|\boldsymbol{\lambda}_2 - \boldsymbol{\lambda}_1\|_2^2$ in the gap.

The function $\boldsymbol{\lambda} \longmapsto \mathcal{L}(\boldsymbol{\lambda}, \boldsymbol{Z})$ does not satisfy this condition, as seen in the right panel for points $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2'$. This function is piecewise-linear, and, for two points on the same line-segment, the first-order Taylor series is exact. However, for points on different line segments, such as $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2$, the first-order Taylor series is a strict underestimation, and we can squeeze in a quadratic correction. Said differently, Equation (4.8) does not hold for all $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_2$ but holds for most $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_2$. In this sense, $\boldsymbol{\lambda} \longmapsto \mathcal{L}(\boldsymbol{\lambda}, \boldsymbol{Z})$ is "approximately" strongly convex.

To make a formal statement, it is more convenient to use a different, equivalent definition of strong convexity. Equation (4.8) is equivalent to the condition

$$(\nabla_{\boldsymbol{\lambda}} f(\boldsymbol{\lambda}_1) - \nabla_{\boldsymbol{\lambda}} f(\boldsymbol{\lambda}_2))^\top (\boldsymbol{\lambda}_1 - \boldsymbol{\lambda}_2) \geq \kappa \|\boldsymbol{\lambda}_1 - \boldsymbol{\lambda}_2\|_2^2 \quad \forall \boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2 \in \mathrm{Dom}(f). \tag{4.9}$$

Lemma D.9 then shows that $\boldsymbol{\lambda} \longmapsto \mathcal{L}(\boldsymbol{\lambda}, \boldsymbol{Z})$ is approximately strongly convex in the sense that, with high probability, there exists a $C > 0$ such that

$$(\nabla_{\boldsymbol{\lambda}} \mathcal{L}(\boldsymbol{\lambda}_1, \boldsymbol{Z}) - \nabla_{\boldsymbol{\lambda}} \mathcal{L}(\boldsymbol{\lambda}_2, \boldsymbol{Z}))^\top (\boldsymbol{\lambda}_1 - \boldsymbol{\lambda}_2) \geq C\|\boldsymbol{\lambda}_1 - \boldsymbol{\lambda}_2\|_2^2 - \frac{\|\boldsymbol{\lambda}_1 - \boldsymbol{\lambda}_2\|_2^{3/2}}{C\sqrt{n}},$$

$$\text{for all } \|\boldsymbol{\lambda}_1\|_1 \leq \lambda_{\max},\ \|\boldsymbol{\lambda}_2\|_1 \leq \lambda_{\max},\ \text{and } \|\boldsymbol{\lambda}_1 - \boldsymbol{\lambda}_2\|_2 \geq \frac{4}{n}, \tag{4.10}$$

where $\lambda_{\max}$ is a dimension independent constant satisfying $\mathbb{E}\|\boldsymbol{\lambda}(\boldsymbol{Z}, \boldsymbol{\theta})\|_1 \leq \lambda_{\max}$.

Equation (4.10) mirrors Equation (4.9). As $n \to \infty$, Equation (4.10) reduces to the analogue of Equation (4.9) for $\|\boldsymbol{\lambda}_1\|_1, \|\boldsymbol{\lambda}_2\|_1 \leq \lambda_{\max}$. Moreover, for $\|\boldsymbol{\lambda}_1 - \boldsymbol{\lambda}_2\|_2 \gg \frac{1}{n}$, the first term on the left of Equation (4.10) dominates the second, so that the right-hand side is essentially a quadratic in $\|\boldsymbol{\lambda}_1 - \boldsymbol{\lambda}_2\|_2$. These relations motivate our terminology approximately strongly convex.

Using this notion of approximate strong convexity, we show in Lemma D.12 that there exists a set $\mathcal{E}_n \subseteq \mathbb{R}^n$ such that $\boldsymbol{Z} \in \mathcal{E}_n$ with high probability, and more importantly, for any $\boldsymbol{z} \in \mathcal{E}_n$, the dual solutions are stable, that is,

$$\|\boldsymbol{\lambda}(\boldsymbol{z}, \boldsymbol{\theta}) - \boldsymbol{\lambda}(\overline{\boldsymbol{z}}, \boldsymbol{\theta})\|_2 \leq \frac{C}{n} \sum_{j=1}^{n} \mathbb{I}\{z_j \neq \overline{z}_j\} \quad \forall \overline{\boldsymbol{z}} \text{ s.t. } \|\boldsymbol{\lambda}(\overline{\boldsymbol{z}})\|_1 \leq \lambda_{\max}. \tag{4.11}$$

Equipped with this dual-stability condition, we can bound the average solution instability and the variance of $D(\boldsymbol{Z})$ as in Theorem 3.5.

However, because the previous stability condition holds with high probability instead of in expectation, we can actually use a modification of McDiarmid's inequality (see Theorem A.5) to prove the following, stronger tail bound.

**Lemma 4.9** (Pointwise Convergence of VGC). *Fix some $\boldsymbol{\theta} \in \Theta$. Under the assumptions of Theorem 4.7, there exists a dimension independent constants $C, n_0$ such that, for any $R > 1$ and $n \geq n_0 e^R$, we have with probability at least $1 - 4\exp(-R)$,*

$$|D(\boldsymbol{Z}, \boldsymbol{\theta}) - \mathbb{E}[D(\boldsymbol{Z}, \boldsymbol{\theta})]| \leq CV^3 \log^2(V) \frac{\log^4(n)\sqrt{n}}{h} \sqrt{R}.$$

We then complete the proof of Theorem 4.7 by (i) invoking a covering argument over $\Theta$ to extend this tail bound to a uniform concentration for the VGC and (ii) again leveraging dual stability to show the in-sample optimism (Equation (4.1a)) concentrates similarly. See Online Appendix D.5 for the details.

**4.2.2. Comparison with Proof Technique to Gupta and Rusmevichientong (2021) and Li and Ye (2022).** Gupta and Rusmevichientong (2021) and Li and Ye (2022) also analyze the behavior of $\boldsymbol{\lambda}(\boldsymbol{Z})$ in the limit as $n \to \infty$. The key to their analysis is observing that the function $\boldsymbol{\lambda} \longmapsto \mathbb{E}[\mathcal{L}(\boldsymbol{\lambda}, \boldsymbol{Z})]$ is strongly convex. Using this property, they prove that

$$\|\boldsymbol{\lambda}(\boldsymbol{Z}) - \boldsymbol{\lambda}^*\|_2 = \tilde{O}_p\left(\frac{1}{\sqrt{n}}\right) \tag{4.12}$$

for some constant $\boldsymbol{\lambda}^*$ that does not depend on the realization of $\boldsymbol{Z}$.

Our analysis via approximate strong convexity takes a different perspective. Specifically, instead of studying the function $\boldsymbol{\lambda} \longmapsto \mathbb{E}[\mathcal{L}(\boldsymbol{\lambda}, \boldsymbol{Z})]$, we study the (random) function $\boldsymbol{\lambda} \longmapsto \mathcal{L}(\boldsymbol{\lambda}, \boldsymbol{Z})$. Although more complex, this analysis permits a tighter characterization of the behavior of the dual variables. In particular, leveraging Equation (4.11), one can prove a statement similar to Equation (4.12) (see Lemma D.17); however, to the best of our knowledge, one cannot easily prove Equation (4.11) given the strong convexity of $\boldsymbol{\lambda} \longmapsto \mathbb{E}[\mathcal{L}(\boldsymbol{\lambda}, \boldsymbol{Z})]$ or Equation (4.12). A simple triangle inequality from Equation (4.12) would suggest the much slower rate $\|\boldsymbol{\lambda}(\boldsymbol{Z}) - \boldsymbol{\lambda}(\overline{\boldsymbol{Z}})\| = O_p\left(\frac{1}{\sqrt{n}}\right)$.

It is an open question whether this tighter analysis might yield improved results for the online linear programming setting studied in Li and Ye (2022).

## 5. Numerical Case Study: Drone-Assisted Emergency Medical Response

We reconsider Example 2.2 using real data from Ontario, Canada. Our data analysis and setup largely mirror Boutilier and Chan (2022); however, unlike that work, our optimization formulation explicitly models response time uncertainty.

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**Data and Setup:** Recall, our formulation decides the location of drone depots ($y_l$) and dispatch rules ($x_{kl}$) where a dispatch rule determines whether to send a drone from depot $l$ to location $k$ when requested. Our objective is to minimize the expected response time to OHCA events. We consider $L = 31$ potential drone depot locations at existing paramedic, fire, and police stations in the Ontario area.

To study the effect of problem dimension on our estimator, we vary the number of OHCA events via simulation similarly to Boutilier and Chan (2022). Specifically, we estimate a (spatial) probability density over Ontario for the occurrence of OHCA events using a kernel density estimator trained on eight years of historical OHCA events. We then simulate $K$ (ranging from 50 to 3,200) events according to this density giving the locations $k$ used in our formulation.

In our case study, $\mu_{kl}$ represents the excess time a drone-assisted response takes over an ambulance-only response. (This objective is typically negative.) We learn these constants by first training a $k$-nearest neighbors regression (kNN) for the historical ambulance response times to nearby OHCAs. (For a sense of scale, the maximum ambulance response time is less than 1,500 seconds = 25 minutes.) We estimate a drone response time based on the (straight-line) distance between $k$ and $l$ assuming an average speed of 27.8 m/s and 20 seconds for take-off and landing (assumptions used in Boutilier and Chan 2022). We then set $\mu_{kl}$ to the difference of the drone time minus the ambulance time. These values are fixed constants throughout our simulations and range from –3,100 to 1,200 seconds.

We take $Z_{kl}$ be normally corrupted predictions of $\mu_{kl}$ where the precisions $\nu_{kl}$ are determined by bootstrapping. Specifically, we take many bootstrap samples of our original historical data set and refit the KNN regression and recompute an estimate of ambulance and drone response times. The precision $\nu_{kl}$ is taken to be the reciprocal of the variance of these bootstrap replicates. Precisions range from $\nu_{\min} = 4 \times 10^{-7}$ to $\nu_{\max} = 2 \times 10^{-4}$.

**Policy Class:** To determine dispatch rules for our case study, we consider the following policies:

$$x(\boldsymbol{Z}, \boldsymbol{W}, (\tau, \mu_0)) \in \arg\min_{x \in \mathcal{X}} \sum_{k=1}^{K} \sum_{j=1}^{L} \Big( \frac{\nu_{jk}}{\nu_{jk} + \tau} Z_{jk} + \frac{\tau}{\nu_{jk} + \tau} (W_{jk} - \mu_0) \Big) x_{jk},$$

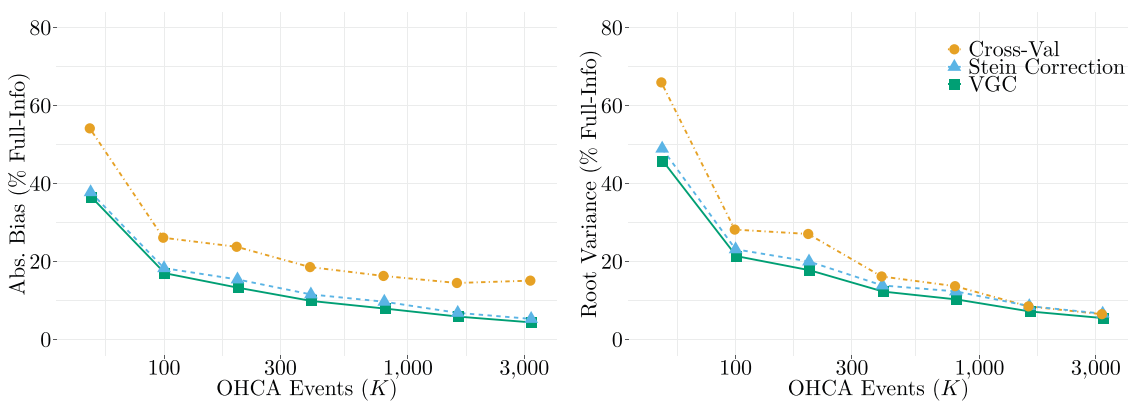
where $W_{jk}$ is the computed drone travel time between facility $j$ and OHCA $k$. Our policy class consists of policies where $\tau \in [0, 100\nu_{\min}]$ and $\mu_0 \in [0, 1500]$. Similar to the mixed-effects policies from Section 2.2, each policy is a weighted average between the SAA policy and a deterministic policy that dispatches to any location whose drone travel-time is at most $\mu_0$.

For the first three experiments, we generate out-of-sample estimates using our VGC, the Stein correction of Gupta and Rusmevichientong (2021), and cross-validation using hold-out (two-fold) cross-validation. We assume that we are given two samples $\boldsymbol{Z}^1, \boldsymbol{Z}^2$ so that $Z_{jk} = \frac{1}{2}(Z^1_{jk} + Z^2_{jk})$. We set $h = n^{-1/6}$ for both the VGC and the Stein correction based on the recommendation of Gupta and Rusmevichientong (2021). In the last experiment, we are given 100 samples $\boldsymbol{Z}^i$ for $i = 1, \ldots, 100$, where $Z_{jk} = \frac{1}{100} \sum_{i=1}^{100} Z^i_{jk}$ and generate out-of-sample estimates for 2-, 5-, 10-, 20-, 50-, and 100-fold cross-validation. For ease of comparison, we present all results as a percentage relative to full-information optimal performance.

## 5.1. Results

In our first experiment, we evaluate the bias and square root variance of each method for the out-of-sample performance of the SAA policy ($\tau = 0$) as $K$, the number of OHCA events grows (Figure 4). As predicted by our theoretical analysis, the quality of the

**Figure 4.** (Color online) Bias and Variance as $K \to \infty$




*Notes.* The two graphs plot the bias and variance of the different out-of-sample performance estimators for the SAA policy. The bias and variance were estimated across 100 simulations for each $K$. Although variance vanishes for all methods as $K$ increases, cross-validation exhibits a nonvanishing bias and is uniformly worse for all $K$.

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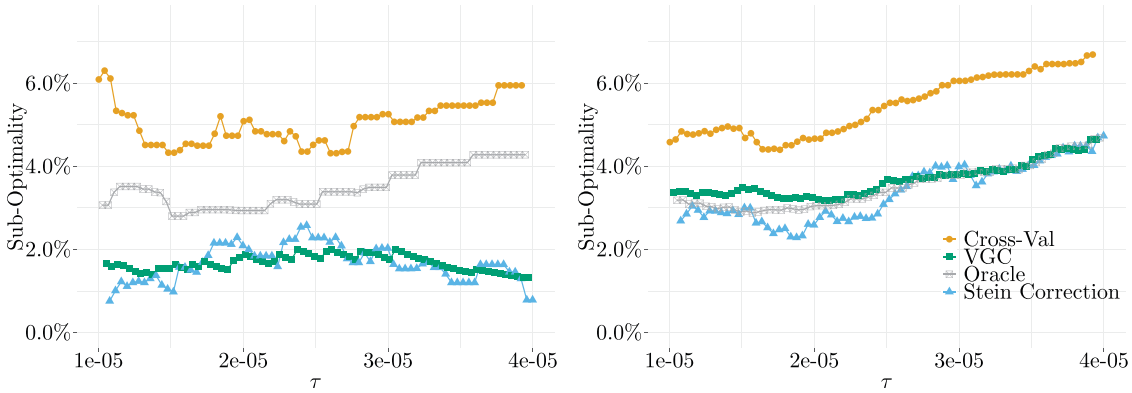
**Figure 5.** (Color online) Estimating Performance Across Policy Class



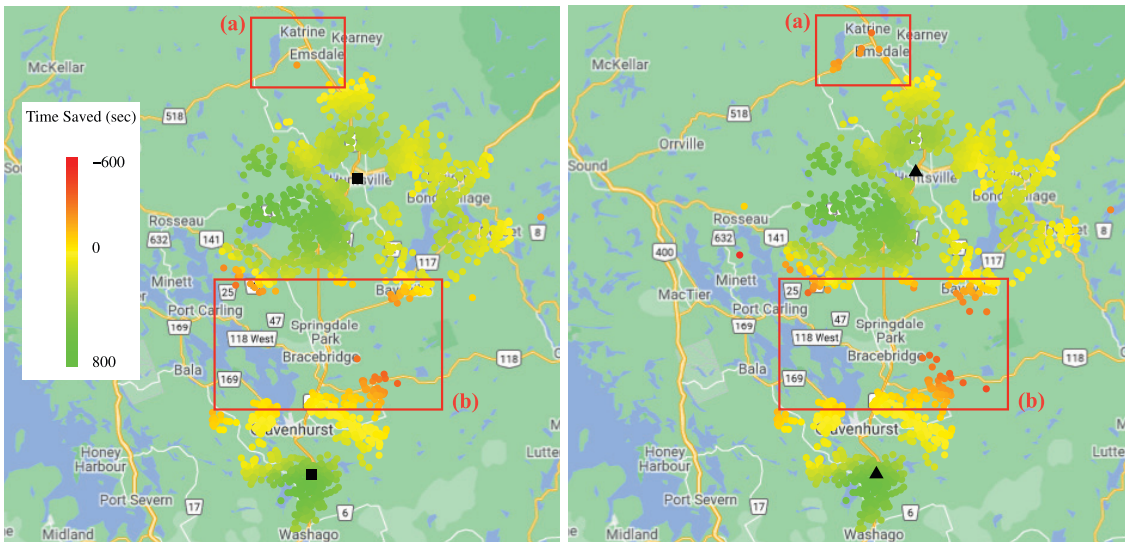

*Notes.* The first graph shows the estimates of out-of-sample performance across the policy class for the parameter $\tau \in [25\nu_{\min}, 100\nu_{\min}]$ and $\mu_0 = 1{,}000$ for one sample path when $K = 400$. The second graph is similar, but for $K = 3{,}200$. Both plots highlight the smoothness of VGC relative to the Stein correction.

out-of-sample estimates improve as we increase the problem size for both the VGC and the Stein correction. However, cross-validation incurs a nonvanishing bias because it only leverages half the data in training.

As a second experiment, in Figure 5, we can observe the quality of the estimators over multiple policies in the policy class. We highlight the smoothness of the VGC as $\tau$ varies. Because, for large $K$, the true performance is quite smooth, the worst-case error of VGC is generally smaller than that of the Stein correction. Although it appears both Stein and VGC systematically overestimate performance, this is an artifact of the particular sample path chosen. By contrast, cross-validation does systematically underestimate performance, because it estimates the performance of a policy trained with half the data, which is necessarily worse.

In our third experiment, we highlight the differences in the policy selected by the VGC estimator and the policy selected by cross-validation. In Figure 6, we plot the routing decisions of each policy and color code them by the true (oracle) amount of time saved. We highlight two regions (labeled (a) and (b)) on the map where drones arrive after the ambulance. We see that in those regions, the cross-validation policy routes to more patients/regions where the drone arrives after the ambulance, thus potentially wasting drone resources and availability. We see in the regions outside of (a) and (b) that routing decisions of the two polices are

**Figure 6.** (Color online) Comparing Policy Decisions




*Notes.* Left (respectively, right) panel shows routing decisions for the policy selected by VGC (respectively, cross-validation). The shade of the dots indicates time-saved relative to an ambulance-only policy computed relative to the ground truth. Although routing is largely similar, regions (a) and (b) highlight some differences where the cross-validation policy makes poorer routing decisions (more time lost dots). The larger black points are drone depots.

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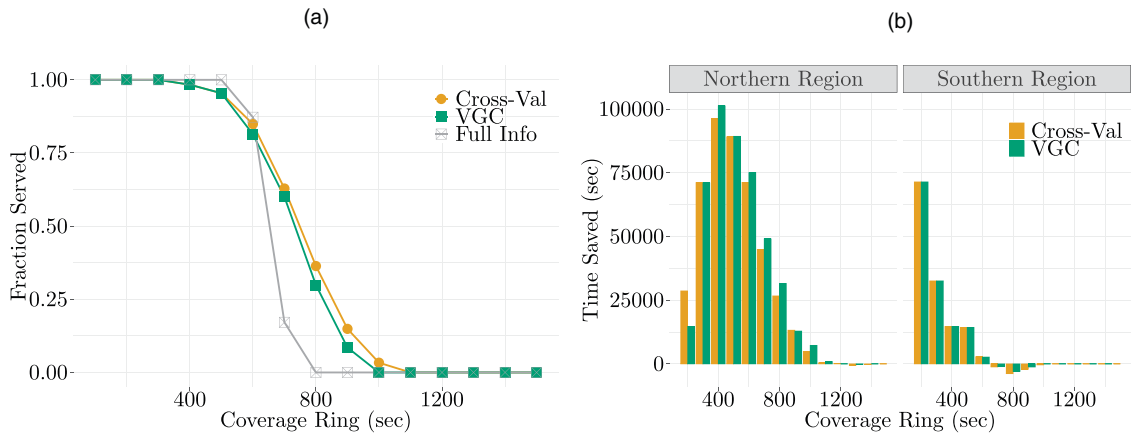
**Figure 7.** (Color online) Estimating Performance Across Policy Class




*Notes.* Each data point in the graph represents the performance metric of each selected policy for a ring-shaped region corresponding to distance in time from a drone depot. (a) Fraction of patients served in each region for the patients serviced by the Southern depot in Figure 6. (b) Time saved by each policy. The plots highlight the performance difference in routing decisions between the two policies.

similar and result in the drone arriving before the ambulance. Additionally, the drone depots in the southern region of the map are the same for both policies whereas the drone depots in the northern region are different.

In Figure 7, we plot key performance metrics for regions organized by their distance from a drone depot for these two policies. Specifically, we group OHCA events into "coverage rings" based on the travel time from the depot to the event. Each ring is of "width" 100 seconds. For example, the 800-second coverage ring corresponds to all OHCA events that are between 701 and 800 seconds away from the drone depot. In Figure 7(a), we restrict attention to the southern region of Figure 6 where both policies have selected the same drone depot so that we can focus on routing decisions. We plot the fraction of patients served for each coverage ring. We see that the policy chosen by VGC is more conservative with routing in comparison with the policy chosen by cross-validation and more closely aligns the full information benchmark.

In Figure 7(b), we compare the time saved between the two policies. We organize the regions into the north and south corresponding to the servicing depots. In the northern region, we see that the VGC policy saves more time in coverage rings further away from the drone depot by sacrificing time saved in closer coverage rings. This difference partially corresponds with the more conservative routing decisions of the VGC, but also can be attributed to the choice of drone depot. We see the VGC policy chooses a depot in less densely populated region that is more centralized overall, while the cross-validation chooses a depot closer to more densely populated regions in terms of OHCA occurrences. In total, we see that the VGC policy saves 1.43% more time in comparison with the cross-validation policy. However, if we break down the time saved with respect to minimum distance from a depot, we see that for patients within 600 seconds of an existing drone depot, the VGC performs less than 1% worse in comparison with the cross-validation policy. However, if we consider patients more than 600 seconds away from existing drone depot, the VGC policy saves 13.8% more time in comparison with cross-validation. We interpret this to mean that both the VGC and cross-validation policies make similar performing depot decisions, but VGC makes significantly better routing decisions, particularly at long distances. Because these long distances are precisely where the imprecision is most crucial, we argue this is a relevant advantage.

Finally, we also compare how higher fold cross-validation performs with respect to VGC. In Figure 8(a), we first show how the cross-validation estimators for three different policies performs in estimation error relative to VGC as we vary the number of folds. For all the policies (where increasing $\tau$ corresponds to lower variance policies), the plot shows the root mean-squared error (MSE) of cross-validation is uniformly larger than VGC over the folds considered. Furthermore, the plot highlights a drawback of cross-validation, which is that it is not even clear how to select the optimal number of folds to minimize the bias variance tradeoff of the cross-validation estimator. In comparison, the VGC with minimal guidance on the choice of the $h$ parameter outperforms common choices of folds for cross-validation such as five-fold cross-validation and leave-one-out cross-validation. In Figure 8(b), we show how the performance of policy evaluation translates to policy learning in regions further away from the depot, where drone decisions are most crucial. As expected, VGC outperforms cross-validation across all choices of folds, and the fold that performs most similarly to VGC, 20-fold,

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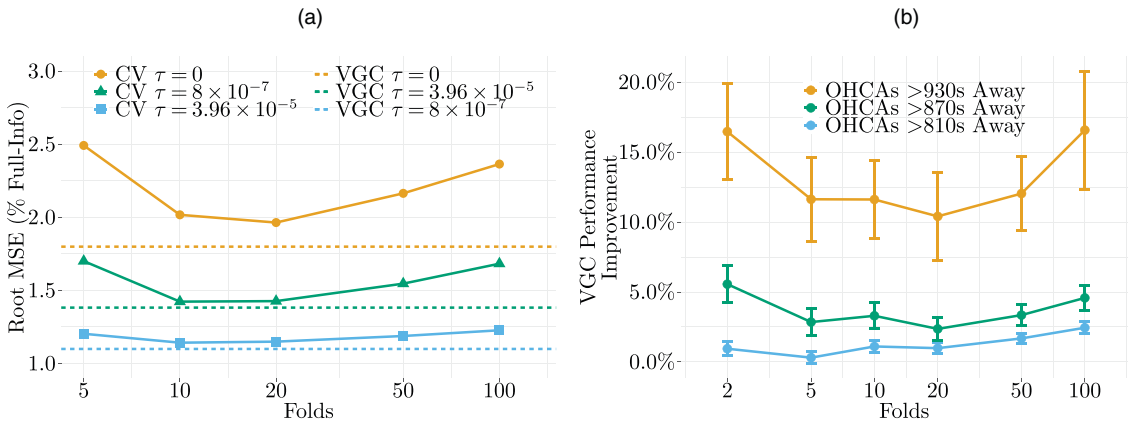
**Figure 8.** (Color online) Varying Cross-Validation Folds




*Notes.* We plot the policy evaluation and learning performance of cross-validation with different folds across 500 simulations. In each simulation, there are 100 samples of $\boldsymbol{Z}^i$. (a) Mean squared error of cross-validation for three different policies compared with the respective VGC estimates represented by the dotted lines. (b) Percent improvement VGC has over cross-validation, so larger bars indicate lower cross-validation performance.

also corresponds to the cross-validation fold with the lowest MSE. This observation suggests with an "ideal" number of folds, cross-validation can perform well in this example, but identifying the right number of folds is nontrivial.

## 6. Conclusion

Motivated by the poor performance of cross-validation in data-driven optimization problems where data are scarce, we propose a new estimator of the out-of-sample performance of an affine plug-in policy. Unlike cross-validation, our estimator avoids sacrificing data and uses all the data when training, making it well-suited to settings with scarce data. We prove that our estimator is nearly unbiased, and for "stable" optimization problems, problems whose optimal solutions do not change too much when the coefficient of a single random component changes, the estimator's variance vanishes. Our notion of stability leads us to consider two special classes of weakly coupled optimization problems: weakly coupled-by-variables and weakly coupled-by-constraints. For each class of problems, we prove an even stronger result and provide high-probability bounds on the error of our estimator that holds uniformly over a policy class. Additionally, in our analysis of optimization problems of weakly coupled-by-constraints, we provide new insight on the stability of the dual solutions. This new insight may provide further insight in problems that leverage the dual solution such as online linear programming.

Our work offers many exciting directions for future research. Our solution approach to the weakly coupled problems exploits the decomposability of the underlying optimization problems. We believe that such an approach can be generalized to other settings. Finally, our analysis strongly leverages the linearity of the affine plug-in policy class; it is an open question if similar debiasing techniques might be developed to handle nonlinear objective functions as well.

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## Endnotes

[1] Following Van der Vaart (2000), we say a sequence of random variables $X_n = O_p(a_n)$ if the sequence $X_n/a_n$ is stochastically bounded, that is, for every $\epsilon > 0$, there exists finite $M > 0$ and finite $N > 0$ such that $\mathbb{P}\{X_n/a_n \geq M\} < \epsilon$, for all $n > N$.

[2] Following Hastie et al. (2020b), we define $SNR = Var(\mu_\pi)/Var(Z_\pi) = \frac{S}{2n}\sum_{j=1}^{n}(\mu_j - \frac{1}{n}\sum_{i=1}^{n}\mu_i)^2$, where $\pi$ is an index drawn uniformly random from one to $n$.

[3] For clarity, this is the same as the performance as $\boldsymbol{x}^{\mathrm{SAA}}(\boldsymbol{Z}^{-2})$ because the data are i.i.d.

[4] See Theorem B.1 in Online Appendix B for a statement of Danskin's theorem.

[5] Here $\sim_d$ denotes equality in distribution.

## References

Aouad A, Elmachtoub AN, Ferreira KJ, McNellis R (2019) Market segmentation trees. Preprint, submitted June 4, https://arxiv.org/abs/1906.01174.

Baardman L, Cohen MC, Panchamgam K, Perakis G, Segev D (2019) Scheduling promotion vehicles to boost profits. *Management Sci.* 65(1):50–70.

Bertsekas DP (1997) Nonlinear programming. *J. Oper. Res. Soc.* 48(3): 334–334.

Bousquet O, Elisseeff A (2000) Algorithmic stability and generalization performance. Leen T, Dietterich T, Tresp V, eds. *Adv. Neural Inform. Processing Systems,* vol. 13 (MIT Press, Cambridge, MA), 196–202.

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Boutilier JJ, Chan TCY (2022) Drone network design for cardiac arrest response. *Manufacturing Service Oper. Management.* 24(5):2407–2424.

Breiman L (1996) Bagging predictors. *Machine Learn.* 24(2):123–140.

Buja A, Hastie T, Tibshirani R (1989) Linear smoothers and additive models. *Ann. Statist.* 17(2):453–510.

Cheskes S, McLeod SL, Nolan M, Snobelen P, Vaillancourt C, Brooks SC, Dainty KN, et al. (2020) Improving access to automated external defibrillators in rural and remote settings: A drone delivery feasibility study. *J. Amer. Heart Assoc.* 9(14):e016687.

Chu J, Leung KHB, Snobelen P, Nevils G, Drennan IA, Cheskes S, Chan TCY (2021) Machine learning-based dispatch of drone-delivered defibrillators for out-of-hospital cardiac arrest. *Resuscitation* 162:120–127.

Cohen MC, Kalas JJ, Perakis G (2021) Promotion optimization for multiple items in supermarkets. *Management Sci.* 67(4):2340–2364.

Cohen MC, Leung N-HZ, Panchamgam K, Perakis G, Smith A (2017) The impact of linear optimization on promotion planning. *Oper. Res.* 65(2):446–468.

Donti P, Amos BJ, Kolter Z (2017) Task-based end-to-end model learning in stochastic optimization. Guyon I, von Luxburg U, Bengio S, Wallach H, Fergus R, Vishwanathan S, Garnett R, eds. *Adv. Neural Inform. Processing Systems*, vol. 30 (Curran Associates Inc., Red Hook, NY), 5490–5500.

Dudík M, Langford J, Li L (2011) Doubly robust policy evaluation and learning. *Proc. 28th Internat. Conf. on Machine Learn.* (Omnipress, Madison, WI), 1097–1104.

Elmachtoub AN, Grigas P (2022) Smart "predict, then optimize". *Management Sci.* 68(1):9–26.

Elmachtoub A, Jason CNL, McNellis R (2020) Decision trees for decision-making under the predict-then-optimize framework. *Proc. Internat. Conf. on Machine Learn.* (PMLR) 2858–2867.

Friedman J, Hastie T, Tibshirani R (2001) *The Elements of Statistical Learning* (Springer, Berlin).

Gamarnik D (2013) Correlation decay method for decision, optimization, and inference in large-scale networks. *Theory Driven by Influential Applications*, Chapter 6, 108–121.

Gelman A, Carlin JB, Stern HS, Rubin DB (2014) *Bayesian Data Analysis*, vol. 2 (Chapman & Hall/CRC, Boca Raton, FL).

Gupta V, Kallus N (2022) Data pooling in stochastic optimization. *Management Sci.* 68(3):1595–1615.

Gupta V, Rusmevichientong P (2021) Small-data, large-scale linear optimization with uncertain objectives. *Management Sci.* 67(1):220–241.

Hastie T, Tibshirani R, Tibshirani R (2020) Best subset, forward stepwise or lasso? Analysis and recommendations based on extensive comparisons. *Statist. Sci.* 35(4):579–592.

Hu Y, Kallus N, Mao X (2022) Fast rates for contextual linear optimization. *Management Sci.* 68(6):4236–4245.

Ignatiadis N, Wager S (2019) Covariate-powered empirical Bayes estimation. Wallach H, Larochelle H, Beygelzimer A, d'Alché-Buc F, Fox E, Garnett R, eds. *Advances in Neural Information Processing Systems*, vol. 32 (Curran Associates Inc., Red Hook, NY) 9620–9632.

Ito S, Yabe A, Fujimaki R (2018) Unbiased objective estimation in predictive optimization. Dy J, Krause A, eds. *Proc. 35th Internat. Conf. on Machine Learn.*, vol. 80, 2176–2185.

Javanmard A, Montanari A (2014) Confidence intervals and hypothesis testing for high-dimensional regression. *J. Machine Learn. Res.* 15(1):2869–2909.

Javanmard A, Montanari A (2018) Debiasing the lasso: Optimal sample size for gaussian designs. *Ann. Statist.* 46(6A):2593–2622.

Javanmard A, Nazerzadeh H, Shao S (2020) Multi-product dynamic pricing in high-dimensions with heterogeneous price sensitivity. *Proc. IEEE Internat. Sympos. on Inform. Theory*, 2652–2657.

Kearns M, Ron D (1999) Algorithmic stability and sanity-check bounds for leave-one-out cross-validation. *Neural Comput.* 11(6):1427–1453.

LeVeque RJ (2007) *Finite Difference Methods for Ordinary and Partial Differential Equations: Steady-State and Time-Dependent Problems* (SIAM, Philadelphia).

Li X, Ye Y (2022) Online linear programming: Dual convergence, new algorithms, and regret bounds. *Oper. Res.* 70(5):2948–2966.

Liu J, Li G, Sen S (2022) Coupled learning enabled stochastic programming with endogenous uncertainty. *Math. Oper. Res.* 47(2):1681–1705.

Liyanage LH, Shanthikumar JG (2005) A practical inventory control policy using operational statistics. *Oper. Res. Lett.* 33(4):341–348.

Pani SR, Abhishek MS (2017) Large-scale advertising portfolio optimization in online marketing. Working paper.

Pollard D (1990) Empirical processes: Theory and applications. *Proc. NSF-CBMS Regional Conf. Series in Probability and Statist.*, vol. 2 (Institute of Mathematical Statistics), i–86.

Rusmevichientong DP, Williamson P (2006) An adaptive algorithm for selecting profitable keywords for search-based advertising services. *Proc. 7th ACM Conf. on Electronic Commerce* (ACM, New York), 260–269.

Sanfridsson J, Sparrevik J, Hollenberg J, Nordberg P, Djärv T, Ringh M, Svensson L, et al. (2019) Drone delivery of an automated external defibrillator: A mixed method simulation study of bystander experience. *Scandinavian J. Trauma Resuscitation Emergency Medicine* 27(1):1–9.

Schmoys D, Wang S (2019) How to solve a linear optimization problem on incentive allocation? Accessed July 21, 2021, https://eng.lyft.com/how-to-solve-a-linear-optimization-problem-on-incentive-allocation-5a8fb5d04db1.

Shalev-Shwartz S, Shamir O, Srebro N, Sridharan K (2010) Learnability, stability and uniform convergence. *J. Machine Learn. Res.* 11:2635–2670.

Shao J (1993) Linear model selection by cross-validation. *J. Amer. Statist. Assoc.* 88(422):486–494.

Smith JE, Winkler RL (2006) The optimizer's curse: Skepticism and postdecision surprise in decision analysis. *Management Sci.* 52(3):311–322.

Van der Vaart AW (2000) *Asymptotic Statistics.* Cambridge Series in Statistical and Probabilistic Mathematics (Cambridge University Press, Cambridge, UK).

Wager S, Athey S (2018) Estimation and inference of heterogeneous treatment effects using random forests. *J. Amer. Statist. Assoc.* 113(523):1228–1242.

Wilder B, Dilkina B, Tambe M (2019a) Melding the data-decisions pipeline: Decision-focused learning for combinatorial optimization. *Proc. Conf. AAAI Artificial Intelligence* 33:1658–1665.

Wilder B, Ewing E, Dilkina B, Tambe M (2019b) End to end learning and optimization on graphs. *Adv. Neural Inform. Processing Systems*, vol. 32 (Curran Associates Inc., Red Hook, NY), 4672–4683.

Zhang C-H, Zhang SS (2014) Confidence intervals for low dimensional parameters in high dimensional linear models. *J. Royal Statist. Soc. Ser. B Statist. Methodology* 76(1):217–242.

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